

China AI-Driven Drug Discovery

Drug Discovery Riding the Next Wave of AI; initiate on Insilico & XtalPi at OW

We initiate coverage of two China AI-driven drug discovery (AIDD) companies, Insilico (3696.HK) and XtalPi (2228.HK), with Overweight (OW) ratings. China's AIDD sector is shifting from business exploration to early commercialization, supported by rising business development (BD) activities and clinical catalysts. Our positive view on China AIDD rests on three pillars: **1)** AIDD has materially improved drug R&D productivity and could further increase success probability through “dry-wet lab” closed loop platforms; **2)** AIDD coupled with China's structural cost and R&D pace and regulatory advantages could make China a more attractive hotbed for applying AIDD compared to overseas; and **3)** The above factors have driven Big Pharma deal flow, clinical milestones, and significant topline growth, providing early validation for AIDD companies. We see a multi-year rerating opportunity, with Insilico and XtalPi offering complementary investment theses. We like Insilico for its high-risk, high reward pipeline built through AIDD and like XtalPi for its diverse businesses led by AIDD services and robotics wet lab. We rate both Insilico and XtalPi OW with PTs of HK\$71 and HK\$10, respectively.

- **AIDD enables a measurable drug R&D unit-economics shift and provides real clinical promise to improve drug success probability.** Closed-loop “dry lab + automated wet lab” platforms move discovery from trial-and-error to iterative, data-recycling execution that compounds a proprietary data flywheel, reducing early drug discovery cost and boosting efficiency. Beyond small molecule and antibody drugs, AIDD is capable of tackling complex and novel drug modalities. In addition, AI platforms are increasingly being integrated into clinical and regulatory workflows to de-risk late-stage drug development.
- **AIDD players based in China are structurally better positioned than some overseas peers because of easy access to China's dense R&D infrastructure,** which helps convert AIDD predictions into validated assets faster and more cheaply than overseas alternatives, anchoring the sector's structural cost advantage. Regulatory tailwinds are accelerating AIDD execution in China, as the NMPA shortens review timelines and builds AI-specific frameworks that lower the friction of moving candidates into human testing. Importantly, the proprietary data, including real-world clinical data from public hospitals in China, allow AIDD companies to leverage high-quality biological, pharmaceutical, and clinical data across many sources.
- **Recent AIDD commercialization through services and BD, plus clinical catalysts, makes the timing to invest in AIDD to be now.** Landmark BD deals with global Big Pharma provide multi-billion-dollar validation, while early-stage asset monetization turns cash-burning biotechs into non-dilutive revenue engines. AIDD platforms are establishing predictable service-driven revenue

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Equity Ratings and Price Targets

Company	Ticker	Mkt Cap (\$ mn)	Price CCY	Price	Rating		Price Target		
					Cur	Prev	Cur	End Date	Prev End Date
XtalPi - H	2228 HK	3,864	HKD	7.04	OW	—	10.00	Dec-27	—
Insilico - H	3696 HK	3,576	HKD	48.44	OW	—	71.00	Dec-27	—

Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates. n/c = no change. All prices as of 23 Jul 26.

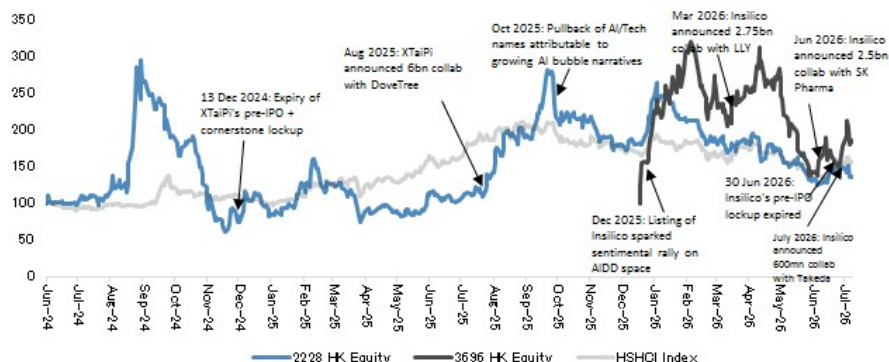
See page 55 for analyst certification and important disclosures, including non-US analyst disclosures.

streams that complement their BD revenue. A wave of top AI and technology companies are moving into AIDD, enforcing the view that the time to work on AIDD is now.

- **We initiate on Insilico and Xtalpi at OW, considering them complementary exposures to AIDD.** Our PT for Insilico Medicine (3696.HK), which has a higher beta than XtalPi, is HK\$71. Our PT for XtalPi (2228.HK) is HK\$10. We suggest owning both with Insilico for catalyst torque and XtalPi for infrastructure ballast. Our valuation uses SOTP based on AI-empowered EV/S on 2027E sales, with DCF as a cross-check. Our PTs imply 19x P/S 2027E for Insilico and 25x for XtalPi, versus overseas AIDD companies that average ~63x. Risks include: clinical trial failure, BD revenue below expectations, US-China drug licensing friction, US sanction risks, AIDD model generalization failure.

Investment Summary

Figure 1: AI + Healthcare sentiment and sizable collab events drive both share prices, and the space is getting more eventful



Source: J.P. Morgan estimates, Bloomberg Finance L.P. Note: Past performance is not an indicator of future results.

Figure 2: AIDD is pivoting towards the execution stage as with increased rollout of projects

Year / Phase	XtalPi	Insilico Medicine
2024	- Mainly fee-for-service R&D - Launched Xtalgazer platform	- ISM001-055 (IPF): positive China Phase IIa topline - Out-licensed MEN2312 (KAT6) to Stemline/Menarini - 5 new PCCs
2025 (Expansion)	- First profitable year (~+201% revenue) - Incubated MTS-004 (METis, PBA): Phase III completed, out-licensed	- ISM001-055 (IPF): Phase IIa data published in <i>Nature Medicine</i> - Out-licensed KIF18A (MEN2501) to Stemline - Launched Science MMAI Gym - ~7 PCCs (2025–early 2026)
2026 & beyond (Execution)	- Pivot to platform licensing / co-development - SIGX2649 (Signet): AACR data, dual INDs - RTX-117 (ReviR), PEP08 (PharmaEngine): Phase I - ~5–6 incubated clinical projects	- Eli Lilly deal up to ~US\$2.75bn - SK Pharma deal up to ~US\$2.5bn - ISM001-055: pivotal Phase III (China) - 10 clinical assets (2 Ph II, 8 Ph I) - Target: 8–10 PCCs/year - 2027+ readouts: ISM5411, ISM3091, ISM5043, ISM3412, ISM6331

Source: J.P. Morgan estimates, Company data.

Figure 3: Clinical catalysts for drug discovered or optimized by AIDD with FIC potential

Company	FIC Asset	Target — Indication	FIC Basis	Stage	Key Catalyst & Timing
Insilico (HK 3696)	Repotertib (ISM001-055)	TNIK — IPF	First TNIK inhibitor in clinic; most advanced AI-discovered/optimized drug globally	Ph III	FPI 2026; topline ~2029
Insilico (HK 3696)	Garutadustat (ISM5411)	Gut-restricted PHD inhibitor — IBD/UC	Potential FIC gut-restricted HIF-PHD inhibitor	Ph IIa	BETHESDA Ph IIa dosing began Jan 2026; topline ~2027
XtalPi (HK 2228) — via Signet	SIGX2649	Pan-TEAD — solid tumors	Potential FIC pan-TEAD inhibitor	IND initiated (Jan 2026)	Preclinical data AACR 2026 (Apr); Ph II entry ~3Q 2026. First clinical readout in ~2027.
Recursion (RXRX)	REC-1245	RBM39 degrader (molecular glue) — solid tumors/lymphoma	Potential FIC RBM39 molecular-glue degrader	Ph 1/2 (DAHLIA)	Prelim ORR readout 1 n2H26
Relay (RLAY)	Zovogalisib (RLY-2608)	PI3Kα (pan-mutant) — HR+/HER2- BC	First allosteric pan-mutant-selective PI3Kα inhibitor to enter clinical development	Ph III	Ph 3 topline readout in 2027-2028E; FDA Breakthrough Therapy (Feb 2026)

Source: Company data.

Figure 4: XtalPi has a diverse set of businesses with robotics as a key feature, while Insilico is a deep drug pipeline play

Feature	XtalPi (QuantumPharm)	Insilico Medicine
Business Model	AI for science drug service provider: Service and co-development platform.	AI-empowered pipeline-deep Biotech: Focused on internal drug development and out-licensing.
Strategic Analogy	The "Automated Kitchen": Provides the robotic tools and AI brains for others to make drugs, while also designs own recipe.	The "Master Recipe Creator": Designs the drug recipe to sell to Big Pharma.
Core Tech Moat	Quantum Physics + Robotics: Large-scale automated wet labs powered by physical algorithms.	Generative AI: End-to-end Pharma.AI platform spanning biology, chemistry, and clinical trials.
Pipeline Strategy	High-Volume Assetization: Economic rights across 500+ collaborative programs.	Clinical Deep-Dive: 20+ proprietary clinical or IND-enabling stage assets.
Best Pipeline Asset	SIGX1094: Co-developed gastric cancer drug; in Phase I trials with FDA Fast Track.	Repotertib (ISM001-055): IPF drug; completed Phase IIa, advancing to Phase III.
Revenue Stream Mix (FY25)	Rmb803mn Drug discovery solutions: 67.0% AI4S intelligent solutions: 33.0%	US\$6.2mn Drug discovery: 44.4% Pipeline development: 42.5% Software solution: 8.7% Other discovery: 4.4%
Sales Growth	Relatively steady: Revenue rose 31%, 53% and 201% YoY in FY23, FY24 and FY25.	More Volatile: Revenue up ~70% in FY23 and FY24 but contracted 34.5% YoY in 2025 due to deal timing.
Gross Margin (FY25)	70%: Dragged down initially by heavy physical lab and robotics CAPEX.	81.5%: Highly lucrative, software-like margins that spike upon milestone execution.
Capital Intensity	Higher: Requires massive footprints for physical wet labs and customized hardware.	Lower: A digital-first, software-heavy approach scaling via external CROs/CDMOs.

Source: J.P. Morgan, Company data.

Figure 5: The Restaurant Analogy: AIDD vs CRO vs Pharma

Analogy	XtalPi	Insilico	CRO*	Biopharma*
Core identity	Kitchen-for-hire turning co-owner	Recipe-inventing kitchen	Pure kitchen-and-chefs-for-hire	Pure recipe-inventing restaurant
Self-own recipe?	Starts with equity in incubatees, now self-developing.	Yes, wholly-owned pipeline first, then collab.	No—always works to order	Yes, by definition
Differentiated AI kitchen?	Yes—quantum AI + robotics	Yes—Pharma.AI	Limited / generic	Usually not
Signature dish (for now)	SIGX1094	Repotertib	None (client keeps IP)	Own lead drug candidate
Value capture	Service fees + equity upside	Upfront + milestones + royalties	Fee-for-service	Drug sales + licensing
Asset intensity	Asset-light via equity	Asset-heavy, high pipeline risk except for collab products	Asset-light, less pipeline risk	Asset-heavy, high pipeline risk

Source: J.P. Morgan estimates, Company data. Note: *Broad business illustration only.

XtalPi and Insilico provide AIDD infrastructure and AIDD drug assets, a bit like AI kitchen. They still need CXO for some preclinical and clinical work & large pharma for future commercialization

Figure 6: AI does substantially boost the economics of drug R&D with time compression from years to months

Company	Metric Category	Traditional Baseline	AI-Empowered Result
XtalPi	Hit-to-Lead Discovery Time	~2.5 years	< 2 months
	400-Molecule Library Synthesis Time	At least 3 weeks	5 days
	Finding Active Target Compounds (Hit Rate)	8.5%	36%
	Crystal Structure Prediction (CSP) Success Rate	86% to 93%	100%
	Crystal Structure Prediction (CSP) Time for a Small Molecule	2 months	2 to 3 weeks
	Binding Pose High-Accuracy Prediction Rate	~30% (commercial software)	~56%
	AI-Driven Reaction Success Rate (Library Synthesis)	Trial-and-Error	>90%
	Chemical Reaction Prediction Accuracy	Human Expert Level	>80%
	LCMS Spectrum Analysis Accuracy	Manual Review	>95%
	X-Buddy R&D Research Cycle Time	Several weeks	30 minutes
Insilico Medicine	Time to Pre-Clinical Candidate (PCC)	4.5 years	12 to 18 months
	Phase I Clinical Success Rate	40% to 65%	80% to 90%
	Overall Probability of Success (Phase I to Approval)	5% to 10%	9% to 18%
	Average Cost per PCC	Traditional high costs	US\$3 million to US\$5 million
	Phase II Trial Outcome Prediction (inClinico)	N/A	0.86 AUC-ROC
	Protein Folding Prediction Accuracy	N/A	84%
	Drug Toxicity Prediction Accuracy	N/A	75%
	TargetPro Novel Target Druggability	Public platform baselines	86.5%
	TargetPro Target Retrieval Rate	Open Targets benchmark	71.6% (2-3x improvement)
	Science MMAI Gym Performance Boost	Baseline Foundation Model	Up to 10X increase

Source: Company reports.

So, our SOTP valuation assigns an AI-premium on top of the traditional sector multiples. For example, XTaiPi's chemical synthesis service was valued with China CRO sector EV/S of ~5x, multiplied by an AI premium of ~4X

Figure 7: Assigning AI premium to sector multiples in XtalPi valuation

XtaiPi	27E Sales	Traditional EV/S	AI Premium R&D efficiency gain	AI-empowered EV/S	Implied EV
SoTP (Rmb mn)					
Intelligence solution	669				13,619
- Robotics	254	3.2x	6.0x	19.2x	4,891
- Chem syn service	414	4.9x	4.3x	21.1x	8,729
Drug discovery	778				14,264
- Collabs	595	8.5x	2.5x	21.3x	12,648
- Subscription/FFS	183	2.1x	4.3x	8.9x	1,616
Total	1,446			19.3x	27,883
Self run assets - NPV					1,315
Net cash/(debt)					6,624
Equity value (Rmb mn)					35,823
Implied P/S (27E)					24.8x
Shares Outstanding (mn)					4,303
Per Share Value (RMB)					8.3
Per Share Value (HK\$)					10

Source: J.P. Morgan estimates, Bloomberg Finance L.P.

While Insilico's drug discovery collaboration business is valued at China biotech EV/S of ~9x, multiplied by ~2x AI-premium

Figure 8: ...In Insilico valuation, the AI premium is based on R&D efficiency gains supported by AI platforms

Insilico	Traditional		AI Premium	AI-empowered	
SoTP (US mn)	27E Sales	EV/S	R&D efficiency gain	EV/S	Implied EV
Drug discovery	271			17.5x	4,737
Collaboration	235	8.8x	2.1x	18.8x	4,417
Subscription	36	2.0x	4.5x	8.9x	320
Software solution & others	9	2.0x	4.5x	8.9x	83
Total	280			17.2x	4,820
Net cash/(debt)					433
Equity value (US mn)					5,253
Implied P/S (27E)					18.8x
Shares Outstanding (mn)					572
Per Share Value (USD)					9.2
Per Share Value (HK\$)					71

Source: J.P. Morgan estimates, Bloomberg Finance L.P.

Our valuation is sensitive to the assigned AI-empowered EV/S and probability-of-success (POS). XTaiPi's valuation will +/- by 4% for 1x +/- of AI-empowered EV/S

Figure 9: XTaiPi valuation is relatively more sensitive to EV/S change

		POS Increment		
		5%	10%	15%
AI-empowered EV/S	17.3x	8.8	8.9	9.0
	18.3x	9.2	9.3	9.4
	19.3x	9.6	10	9.8
	20.3x	10.0	10.1	10.1
	21.3x	10.4	10.5	10.5

		POS Increment		
		5%	10%	15%
AI-empowered EV/S	17.3x	-9%	-8%	-7%
	18.3x	-5%	-4%	-3%
	19.3x	-1%	-	1%
	20.3x	3%	4%	5%
	21.3x	7%	8%	9%

Source: J.P. Morgan estimates.

Figure 10: Insilico valuation, based more on its deep drug pipeline, is more sensitive to our PoS (probabilities of success) assumptions

		POS Increment		
		10%	15%	20%
AI-empowered EV/S	15.2x	60.9	63.5	66.1
	16.2x	64.6	67.3	70.1
	17.2x	68.2	71	74.0
	18.2x	71.8	74.9	78.0
	19.2x	75.5	78.7	82.0

		POS Increment		
		10%	15%	20%
AI-empowered EV/S	15.2x	-14%	-11%	-7%
	16.2x	-9%	-5%	-1%
	17.2x	-4%	-	4%
	18.2x	1%	5%	10%
	19.2x	6%	11%	15%

Source: J.P. Morgan estimates.

Our implied 2027E P/S numbers for XTaiPi and Insilico are 25x and 19x respectively, which are in fact conservative compared to overseas AIDD averaging ~66x; let alone overseas AIDD companies do not benefit from the efficiency of China's drug making infrastructure

Figure 11: AIDD sector comp sheet

	Ticker	Price (LC) 7/23/2026	Share px perf (%)					ATV 90D (US\$m)	Mkt Cap (US\$b)	EV (US\$b)	EV/S (x)			P/S (x)			Sales CAGR		P/E (x)		
			1M	6M	12M	36M	YTD				26E	27E	28E	26E	27E	28E	26-28E	26E	27E	28E	
China AI Drug Discovery																					
	XTALPH HOLDINGS	2228 HK	7.0	-1%	-48%	19%	33%	-26%	60.2	3.9	2.9	19.4	14.5	10.8	25.6	19.1	14.3	34%	-	216.5	79.8
	INSILICO MEDICIN	3696 HK	48.4	13%	-23%	101%	101%	30%	52.9	3.6	3.2	18.0	12.8	10.3	20.2	14.4	11.5	33%	93.5	23.3	12.3
	Sub-sector average			6%	-35%	60%	67%	2%	56.6		3.0	18.7	13.6	10.5	22.9	16.7	12.9	33%	NM	119.9	46.0
Overseas AIDD																					
	RECURSION PHAR-A	RXRX US	3.0	-6%	-37%	-54%	-81%	-27%	67.4	1.6	1.0	15.1	8.8	4.7	23.9	14.0	7.4	80%	-	-	-
	ABCELLERA BIOLOG	ABCL US	5.7	2%	35%	15%	-27%	68%	45.8	1.7	1.4	43.7	45.1	41.7	54.8	56.6	52.2	2%	-	-	-
	SCHRODINGER INC	SDGR US	15.0	0%	-8%	-35%	-72%	-16%	19.8	1.1	0.8	3.6	3.3	2.6	4.8	4.4	3.6	16%	-	-	-
	GENERATE BIOMEDI	GENB US	13.9	-10%	-13%	-13%	-13%	-13%	11.5	1.8	1.3	42.8	46.9	34.6	57.4	62.7	46.3	11%	-	-	-
	RELAY THERAPEUTI	RLAY US	19.3	10%	159%	408%	56%	128%	64.6	4.1	3.5	255.5	142.8	82.8	300.3	167.9	97.3	76%	-	-	-
	ABSCI CORP	ABSI US	8.2	11%	183%	134%	242%	136%	45.5	1.4	1.3	345.2	70.5	40.0	378.2	77.3	43.8	194%	-	-	-
	Category average											117.6	62.9	34.4	136.6	63.8	41.8	63%	NM	NM	NM
China Biotech/Pharma with Outlicensing deals																					
	INNOVENT BIOLOGI	1801 HK	91.0	19%	9%	3%	177%	19%	157.9	20.2	18.0	6.8	5.4	4.6	7.6	6.1	5.1	21%	63.0	38.7	27.4
	SICHUAN KELUN-H	6990 HK	519.5	31%	21%	39%	597%	32%	63.0	16.4	15.8	39.5	23.5	15.7	41.1	24.5	16.3	59%	-	150.7	59.7
	AKESO INC	9926 HK	102.5	20%	-12%	-29%	176%	-9%	182.1	12.0	11.4	15.4	10.4	7.4	16.3	11.0	7.9	44%	NM	75.9	32.7
	ASCENTAGE PHARMA	6855 HK	37.1	18%	-27%	-50%	65%	-29%	13.0	1.8	1.7	10.6	5.5	4.9	11.0	5.8	5.1	47%	-	NM	168.6
	HANSON PHARMACEU	3692 HK	32.5	10%	-19%	-6%	164%	-10%	48.3	25.2	20.6	8.3	7.6	6.6	10.1	9.2	8.1	12%	28.6	26.1	22.7
	REMEGEN CO LTD-H	9995 HK	87.2	30%	-3%	31%	100%	21%	50.1	8.9	8.8	8.5	12.8	9.9	8.6	12.9	10.0	-7%	13.7	46.3	26.8
	BEONE MEDICINE-H	6180 HK	193.6	20%	-5%	7%	64%	8%	115.1	39.8	36.2	5.6	4.8	4.3	6.2	5.3	4.7	15%	50.3	35.6	27.1
	SHANGHAI JUNSH-H	1877 HK	19.2	18%	-15%	-21%	-23%	-10%	10.4	4.9	5.1	10.4	8.2	6.6	9.9	7.8	6.3	26%	-	103.8	27.2
	JIANGSU HENGRU-H	1276 HK	57.8	3%	-21%	-20%	31%	-19%	32.5	53.6	47.8	9.0	7.8	6.7	10.1	8.7	7.6	15%	34.2	29.1	24.9
	ZAI LAB LTD	9688 HK	16.0	15%	12%	-46%	-32%	17%	20.4	2.3	1.9	4.3	3.3	2.5	5.3	4.1	3.0	32%	-	-	-
	AB&B BIO-TECH-H	2627 HK	29.3	0%	-43%	127%	127%	-27%	NM	1.5	1.6	13.4	8.6	-	12.5	8.0	-	NM	-	78.7	-
	HBM HOLDINGS LTD	2142 HK	12.4	18%	1%	44%	675%	2%	5.8	1.4	1.1	4.7	5.0	4.4	6.1	6.6	5.8	3%	6.5	5.1	3.1
	INNOCARE PHARM-H	9909 HK	14.2	32%	20%	-20%	90%	15%	14.6	3.8	3.0	8.9	7.0	5.6	11.3	8.9	7.1	26%	105.3	86.0	43.9
	Category average											11.2	8.5	6.6	12.0	9.2	7.2	24%	43.1	61.8	42.2
China SaaS																					
	KINGDEE INTL SFT	268 HK	7.1	15%	-50%	-57%	-35%	-46%	50.1	3.2	2.7	2.3	2.0	1.8	2.8	2.4	2.1	14%	52.1	29.7	21.3
	CRRC TIMES ELEC	3898 HK	33.7	-15%	-19%	2%	10%	-10%	28.1	8.1	8.9	1.9	1.7	1.5	1.7	1.5	1.3	13%	8.8	7.8	7.0
	WEIMOB INC	2013 HK	1.2	-6%	-50%	-42%	-68%	-36%	8.4	0.6	0.7	2.5	2.2	2.1	2.3	2.1	2.0	9%	84.2	31.6	20.2
	YONYOU NETWORK-A	600588 CH	9.5	-2%	-41%	-33%	-48%	-29%	83.3	4.8	5.2	3.6	3.3	2.9	3.3	3.0	2.7	10%	-	131.3	52.8
	KINGSOFT CORP	3888 HK	24.3	12%	-20%	-30%	-25%	-14%	49.9	4.3	1.6	1.0	0.9	0.8	2.8	2.5	2.2	12%	17.5	19.7	15.0
	Category average											2.3	2.0	1.8	2.6	2.3	2.1	12%	40.7	44.0	23.3
China CRO																					
	WUAPPTC CO LT-H	2359 HK	158.5	21%	39%	70%	127%	61%	115.5	56.5	52.7	6.7	5.7	5.0	7.1	6.1	5.4	15%	21.9	18.7	15.9
	PHARMARON BEL-H	3759 HK	23.7	53%	10%	22%	36%	21%	18.9	9.6	10.3	4.2	3.7	3.2	4.0	3.4	3.0	16%	17.8	14.2	11.2
	HANGZHOU TIGER-H	3347 HK	37.7	22%	-26%	-22%	-17%	-11%	9.7	5.9	6.3	5.6	4.9	4.3	5.1	4.5	4.0	14%	NM	24.9	20.0
	ASYMCHM LABOR-H	6821 HK	118.0	22%	38%	22%	23%	66%	10.6	8.5	7.4	6.2	5.3	4.5	7.1	6.0	5.2	17%	26.1	20.0	17.3
	Category average											5.7	4.9	4.3	5.8	5.0	4.4	15%	21.9	19.4	16.1
China Humanoid/Service robots																					
	UBTECH ROBOTIC-H	6880 HK	81.6	-17%	-43%	-9%	-9%	-35%	125.3	5.3	4.8	8.4	5.5	3.6	9.3	6.2	4.0	53%	-	288.6	50.4
	BEIJING ROBORO-A	688169 CH	100.3	6%	-34%	-38%	-21%	-34%	128.3	3.8	2.9	0.9	0.7	0.6	1.1	1.0	0.8	19%	13.4	10.4	8.6
	Category average											4.6	3.1	2.1	5.2	3.6	2.4	36%	13.4	149.5	29.5
Global Robotics																					
	SYMBOTIC INC	SYM US	42.5	10%	-32%	-21%	-3%	-29%	100.5	25.6	24.0	8.6	6.6	5.2	9.1	7.1	5.6	28%	116.0	57.1	32.3
	FANUC CORP	6954 JP	6,938.0	-7%	6%	62%	47%	14%	325.8	41.7	37.1	7.2	6.4	6.1	8.1	7.2	6.8	9%	40.2	32.4	30.2
	TESLA INC	TSLA US	374.0	-2%	-17%	12%	44%	-17%	19,413.0	1,404.7	1,371.2	13.1	11.4	9.8	13.4	11.7	10.0	16%	197.3	156.0	110.0
	Category average											9.6	8.2	7.0	10.2	8.7	7.5	0.2	117.8	81.8	57.5
China AI models																					
	Z AI CO LTD	2513 HK	1,177.0	-46%	513%	913%	913%	913%	636.2	69.9	69.7	120.1	41.1	19.4	120.5	41.2	19.4	149%	-	-	-
	MINIMAX GROUP IN	100 HK	197.6	-62%	-47%	20%	20%	20%	192.6	8.8	7.9	22.8	8.3	3.8	25.4	9.3	4.2	146%	-	-	-
	Category average											71.5	24.7	11.6	73.0	25.2	11.8	1.5	NM	NM	NM

Source: Bloomberg Finance L.P. Priced as of 23 Jul 2026.

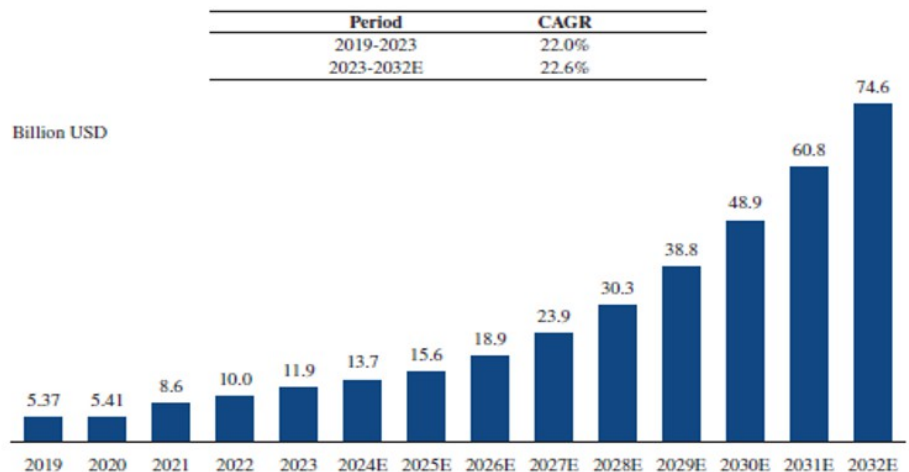
AIDD continues to revolutionize drug discovery and development

Ever expanding AIDD market

The China AIDD sector is decisively transitioning from algorithmic promise to industrial execution. We think the core investment thesis rests on: **1) A paradigm shift in R&D unit economics**, driven by powerful AIDD platforms/tools. More importantly, such AIDD platforms are becoming stronger because of closed-loop "dry-wet" lab systems (Pillar 1); **2) AIDD is uniquely anchored by China's unmatched supply chain density in drug R&D** (Pillar 2); and **3) AIDD continues to be validated as de-risked by Big Pharma milestone capital** (Pillar 3).

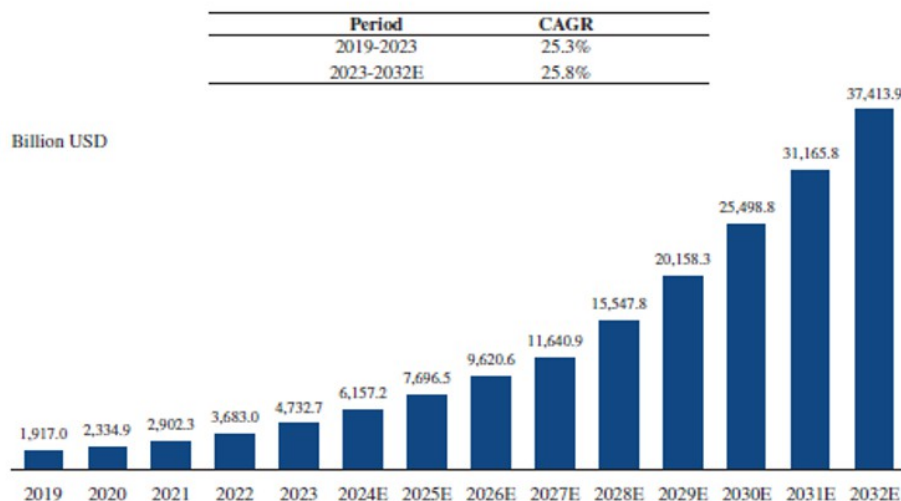
- **The AIDD sector targets a rapidly expanding market ecosystem, with core AI-empowered pharmaceutical R&D spending to thrive, alongside a substantially broader generative AI opportunity.** Driven by the pharmaceutical industry's urgent need to compress drug development cycles, lower costs and discover biologically sound new targets, global spending on AI-empowered drug R&D is forecast by F&S to grow at a 22.6% compound annual growth rate (CAGR) from US\$11.9 billion in 2023 to US\$74.6 billion by 2032. Furthermore, this sector is supported by the broader global generative AI market, which, according to F&S, is expanding at a 25.8% CAGR to reach a notable US\$37.4 trillion by 2032. This vast technological expansion not only provides the foundational infrastructure for pharmaceutical platforms but also enables highly lucrative cross-sector commercialization into adjacent markets like advanced materials and agriculture.

Figure 12: Global AI-empowered drug R&D expense market is expected to grow at 23% CAGR up to 2032E



Source: Frost & Sullivan.

Figure 13: Drug R&D is just one area AI is helping transform. The global generative AI market is expected to grow at a 26% CAGR up to 2032E



Source: Frost & Sullivan.

- Investment activity across the AI drug discovery landscape has been remarkably robust over 2021–2026, underscoring the sector's sizeable market potential.** Marquee venture rounds, Nasdaq IPOs, and multi-billion-dollar pharma partnerships have drawn sustained conviction from tier-one crossover investors, sovereign wealth funds, and strategic corporate backers including Nvidia, Alphabet, Pfizer, and Amgen. The breadth of financing sources, the scale of individual rounds, and the escalating deal values across venture, IPO, and pharma-collaboration channels show that AIDD is transitioning from a speculative theme into a durable, well-capitalized industry with material commercial upside.

Figure 14: Globally, a lot of funding has flowed into the AIDD space recently, indicating strong market opportunities

Date	Company	Event / Round	Amount Raised
14-Jul-26	Chai Discovery	Series C	\$400M
May-26	METiS TechBio (7666.HK)	IPO (HKEX, Ch.18C)	~\$270M (HK\$2.1B)
Feb-26	Generate:Biomedicines	IPO	\$400M
Dec-25	Insilico Medicine	IPO (HKEX)	\$293M
Dec-25	Chai Discovery	Series B	\$130M
Sep-25	Enveda Biosciences	Series D	\$150M
Jul-25	Absci	Public offering	\$50M
Jul-25	METiS TechBio	Series C	~\$49M (RMB 350M)
May-25	Pathos AI	Series D	\$365M
Mar-25	Isomorphic Labs	Growth equity	\$600M
Feb-25	XtalPi	Post-IPO placement	\$268M
Nov-24	Recursion x Exscientia	Merger (closed)	\$688M
Sep-24	BioAge Labs	IPO	\$198M
Jun-24	BioMap	Strategic round	Undisc.
Apr-24	Xaira Therapeutics	Series A	\$1.0B
Jul-23	Recursion	PIPE	\$50M
Apr-22	Owkin	Series B-1	\$80M
Apr-22	METiS Therapeutics	Series B	\$150M
Apr-22	BenevolentAI	SPAC	\$242M
Oct-21	Exscientia	IPO	\$465M
Mar-21	insitro	Series C	\$400M
Mar-21	Valo Health	Series B	\$300M
Dec-20	AbCellera	IPO	\$483M

Source: Company reports.

The future of AIDD platforms/tools lies in closed loop, underlying LLM, and agent use with strong pharmaceutical knowledge

The integration of AI across the pharmaceutical value chain has produced a distinct ecosystem of AIDD platforms and tools but we don't see a single winning AIDD platform yet. These tools fundamentally shift research from a trial-and-error physical paradigm to a predictive, data-driven methodology, addressing distinct bottlenecks ranging from initial biological target discovery to molecular design and clinical trial forecasting. Based on our observations, we think AIDD tooling has converged around three “stacks” that differ mainly in (i) what is treated as ground truth (physics/chemistry/biology vs. learned representations) and (ii) where AI is used (speed-up, surrogate modeling, or end-to-end generation/reasoning). **1)** Physics/pharmacology/biology/physiology models with AI used to accelerate computation. These platforms keep classical mechanistic formalisms (molecular dynamics [MD], quantum mechanics, PK/PD models, systems biology) as the core, and add machine learning (ML) for speed, scoring, or parameterization. Typical use: virtual screening triage, faster free-energy approximations, improved scoring functions, or adaptive sampling in molecular dynamics. Examples include: Schrödinger-style docking/FEP workflows; OpenMM-based MD ecosystems with ML potentials/surrogates, and Simcyp physiologically based

pharmacokinetic (PBPK) modeling tool; **2)** Modern AI architectures (transformers/diffusion/graph nets) grounded in structural or biological data. Deep learning is the primary engine, trained on biological data often with physics-inspired constraints (equivariance, energy-based objectives) or structure supervision. Well-known examples include AlphaFold 4, RoseTTAFold, graph neural network property predictors and multitask models for ADMET (absorption, distribution, metabolism, excretion, and toxicity); diffusion/transformer molecular generators used for scaffold hopping and constrained optimization; and **3)** Large language models (LLMs) post-trained on pharmaceutical data for data analysis, reasoning, planning, and knowledge workflows. These systems are used in medicinal chemistry, pharmacology, CMC, and literature/patent corpora, sometimes connected to tools (search, ELNs, assay databases) for grounded answers. Typical use: hypothesis generation, SAR summarization, protocol drafting, target/indication landscape reviews, and decision support. Examples include domain-adapted LLM assistants built on top of general LLMs; chemistry-aware LLM toolchains that call retrosynthesis/property predictors; literature/patent copilots used by R&D teams. In addition, various agent systems are built on top of LLMs to further facilitate the use of all three classes of AIDD tools. Overall, the development of AIDD is moving toward hybrid stacks: domain knowledge such as physics for correctness, deep models for priors and speed, and LLMs for orchestration and human-facing reasoning.

Figure 15: A few well-known AIDD platforms already find good applications in real-world scenarios

AI Tool / Platform	Developer / Company	Primary Function in Drug Discovery
AlphaFold (2 & 3)	Google DeepMind / Isomorphic Labs	3D protein structure prediction and multi-molecule complex interactions.
PandaOmics	Insilico Medicine	Multi-omics data analysis for novel disease target and biomarker discovery.
Chemistry42	Insilico Medicine	Generative AI for <i>de novo</i> small molecule design, ADMET profiling, and retrosynthesis.
AtomNet	Atomwise	Deep convolutional neural networks for high-throughput virtual screening and molecule-target docking.
Glide & FEP+	Schrödinger	Physics-based molecular simulation and highly accurate binding free energy calculations.
ID4Inno (ID4Gibbs/ID4Idea)	XtalPi	Quantum physics-based computation combined with generative AI for small molecule virtual screening and design.
inClinico	Insilico Medicine	Generative AI for predicting clinical trial outcomes and optimizing Phase II to Phase III transitions.
XtalFold / XenProT	XtalPi	Antigen-antibody complex structure prediction and generative design for large molecule biologics.
IsoDDE	Isomorphic Labs	Proprietary drug design engine integrating structures, pockets, and affinities into compound ranking.

Source: Company data.

- **Deep learning structural prediction models provide the foundational three-dimensional blueprints required for drugs to target previously "undruggable" proteins.** Tools like AlphaFold utilize advanced diffusion-based architectures to translate one-dimensional genetic sequences into highly accurate 3D models of isolated proteins, and its newer iterations can predict interactions within complex biological systems involving proteins, nucleic acids, and small molecule ligands. This allows scientists to visually hypothesize binding pockets and interactions computationally before initiating costly physical experiments.
- **Biocomputational platforms function as advanced biological search engines to rapidly identify and validate disease-driving targets.** Applications such as

Insilico Medicine's PandaOmics deploy natural language processing and machine learning to analyze massive datasets spanning clinical trials, scientific text, and multi-omics (genomic, transcriptomic, and proteomic) data. By recognizing complex biological patterns, these tools can pinpoint highly specific, novel therapeutic targets and biomarkers that traditional human-led research might entirely overlook.

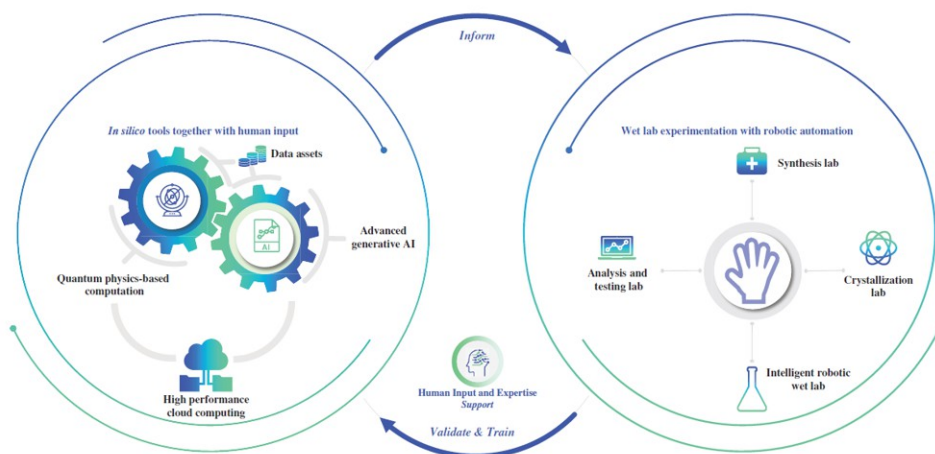
- **Generative chemistry and physics-based simulation engines automate the creation and refinement of entirely novel molecules.** Platforms like Chemistry42, AtomNet, and ID4Inno do not merely search existing chemical libraries; they act as automated molecular architects that invent new chemical structures optimized for specific targets. They seamlessly integrate quantum physics-based calculations to predict binding affinities (such as Schrödinger's FEP+ or XtalPi's XFEP) and utilize predictive algorithms to proactively forecast critical drug-like properties, such as toxicity and metabolic stability (ADMET), ensuring only the most viable candidates advance to physical synthesis.
- **Predictive clinical software optimizes trial architecture to significantly reduce the risk of late-stage development failures.** Tools like inClinico analyze historical trial data, electronic health records, and biological markers to identify patient subgroups most likely to respond to a given therapy. By dynamically simulating patient responses and suggesting adaptive trial protocols, these AI applications help pharmaceutical companies design highly targeted, efficient clinical trials that improve the probability of regulatory success.
- **Multi-agent AI algorithms and advanced robotics reduce manual trial-and-error, driving unprecedented R&D throughput.** Platforms like XtalPi deploy comprehensive multi-agent architectures—such as Agentic Genius, MolAgent, and X-Buddy—that function as intelligent project managers to autonomously orchestrate over 10,000 compound synthesis experiments each week. In the physical laboratory, proprietary hardware overcomes micro-scale material bottlenecks, allowing robotic workstations to autonomously execute up to 1,500 parallel experiments daily. Insilico deploys agentic AI as a "digital research workforce" of 660 specialized agents (as of mid-2024) that autonomously automate multi-step scientific workflows and heavy documentation, cutting the manual burden of literature review and data interpretation. Its flagship DORA (Draft Outline Research Assistant), housed in the Science42 platform, is a multi-agent system that generates hypotheses, interprets data, and drafts academic papers and regulatory documents by synthesizing vast scientific literature into contextual summaries.

The 3 Key Pillars of China's AIDD

Pillar 1: AIDD Could Constantly Improve R&D Efficiency via “Dry-Wet Lab” Closed Loop

"What can AI solve for drugs?" The core value proposition of AIDD is a step-change in R&D unit economics. Most importantly, the “dry-wet lab” closed loops (AI + automation + wet-lab execution) utilized by major AIDD players in China are substantially compressing discovery timelines and lowering capital burn in drug discovery processes. The “dry lab” utilizes generative AI to virtually design molecules and predict their biological properties. These digital predictions are then sent directly to the “wet lab” – a highly automated physical laboratory equipped with intelligent robotics - which rapidly synthesizes the compounds and tests them in the real world. The resulting experimental data is fed back into the dry lab to validate the initial prediction, effectively retraining the AI models to become smarter for the next round of drug design.

Figure 16: The “dry-wet lab” loop system enables AI models to continue to improve, forming an ever deeper moat



Source: Company reports.

- **The integration of AI drastically compresses traditional "Hit-to-Lead" timelines and establishes a significant cost arbitrage.** Insilico Medicine utilizes its end-to-end Pharma.AI platform to shorten the progression of a drug candidate from target discovery to Pre-Clinical Candidate (PCC) nomination down to an average of just 12 to 18 months, bypassing the traditional industry average of 4.5 years. By systematically lowering early-stage failure rates and avoiding the costly physical synthesis of redundant molecules, Insilico demonstrates an unprecedented average cost of only US\$3 million to US\$5 million to successfully generate a single PCC.
- **AIDD platforms function as high-volume asset factories capable of tackling complex novel modalities.** Driven by improved time and capital efficiency, companies like Insilico have scaled output to successfully execute 29 PCC business development deals, while METiS TechBio (incubated by XTalPi) uses its AI-empowered platforms to identify optimal formulations in less than three months compared to the traditional one-to-two-year cycle. Expanding beyond traditional

small molecules, AIDD pioneers are deploying their core computational engines to systematically engineer entirely new therapeutic classes. XtalPi has launched dedicated platforms such as XGlue to enhance the discovery of precise protein interactions for molecular glues, and PepiX, which integrates generative design with a trillion-entry virtual library to discover novel peptides, including advanced blood-brain barrier-penetrating molecules. Furthermore, XtalPi optimizes oligonucleotides and mRNA sequences through its Kodexia platform, utilizing proprietary algorithms to significantly improve mRNA sequence expression levels and stability, proving that these universal AI architectures can accelerate innovation across the most challenging modern drug modalities.

Figure 17: Emerging AI tools support robust asset generation in different modalities and diseases

Company	Target / Asset	Modality	AI Platform	Why AI Was Essential
Insilico Medicine	TNIK (TRAF2- and NCK-Interacting Kinase)	First-in-class kinase inhibitor for idiopathic pulmonary fibrosis	Pharma.AI — PandaOmics + Chemistry42 (molecule generation)	Its fibrosis role was unknown; found only by AI mining of multi-omics networks
Insilico Medicine	VAV1 (Vav Guanine Nucleotide Exchange Factor 1)	Molecular glue degrader	Pharma.AI — Chemistry42	Glue-induced new protein interactions are invisible to standard screening
Insilico Medicine	STAT6 (Signal Transducer and Activator of Transcription 6)	Oral protein degrader (PROTAC)	Pharma.AI — Chemistry42	Three-body complex shapes are too complex for manual design
Insilico Medicine	Pan-KRAS (Kirsten Rat Sarcoma viral oncogene)	Inhibitor hitting all KRAS forms	Pharma.AI — Chemistry42	Multi-state binding plus selectivity over close cousins
XtalPi	Undruggable target proteins	Molecular glue degraders	XtalPi molecular glue platform	Million-entry virtual libraries find binding surfaces too vast to test by hand
XtalPi	Intrinsically disordered proteins	Oral and brain-penetrating peptides	PepiX	These proteins lack stable pockets; design needs many traits balanced at once
XtalPi	Messenger RNA / small interfering RNA	Nucleic acid therapies	Kodexia (mRNA2vec)	Sequence and modification options span a space too large to search manually
XtalPi	Structure-predicted protein targets	Biologics and bispecific antibodies	XtalFold / XenProT	Predicts antibody complex shapes and fixes chain mispairing beyond manual iteration
Recursion(US)	Cell-image-derived cancer and rare-disease targets	Small molecules	Recursion OS	Reads patterns in millions of cell images humans cannot interpret
Isomorphic Labs (UK)	Structure-predicted protein targets	Small molecules	AlphaFold-based platform	Predicts 3D shapes for proteins with no solved crystal structure
Generate Biomedicines(US)	Newly designed proteins	Generative protein drugs	GENERATE platform	Invents protein shapes that do not exist in nature
Insitro (US)	Genetics-derived metabolic and brain targets	Small molecules	insitro Human platform	Links genes to disease across very large patient datasets

Source: Company reports.

- **A good example of how to use AIDD tools for drug target discovery: Insilico's April 2026 scientific paper ([link](#)) shows that its AI target-discovery framework meaningfully lowers early-stage R&D cost by improving the odds that scientists pursue the right targets from the start.** The study introduces TargetPro, a disease-specific model built on the PandaOmics engine that fuses 22 omics and text signals across 38 diseases, paired with TargetBench 1.0, a standardized benchmarking system that validates predictions against reproducible criteria. The framework retrieved known clinical targets with 71.6% precision-at-top-K—1.7x to 5.5x better than leading LLM-based methods—and produced novel candidates that were overwhelmingly "development-ready": 95.7% already have 3D crystal structures,

86.5% are druggable with clinical evidence, and 46% are tied to approved drugs elsewhere, opening repurposing paths. From an R&D economics standpoint, this matters because target selection is where most drug programs fail; by front-loading validation and screening out low-probability targets before costly wet-lab, hit-discovery, and preclinical work begins, the platform reduces expensive late-stage attrition and shortens the discovery cycle, translating directly into lower cost-per-viable-target and more efficient capital deployment across the pipeline.

- **Beyond preclinical asset generation, AI platforms are increasingly being integrated into clinical and regulatory workflows to de-risk late-stage development.** Representative use cases include patient matching, trial simulation and synthetic control arms/digital twins to reduce enrollment needs and speed execution, plus automated drafting/formatting of regulatory documentation and responses to authority questions.

Figure 18: Not just for early-stage drug discovery, AI tools are heavily applied in other important processes across the drug development value chain

Stage	AI Involvement	Improvement / Benefits	Genre of AI Model Utilized
Drug Discovery & Preclinical	Target and structure prediction, novel molecule generation, ADMET profiling, and predictive toxicology (e.g., XTAiPi and Insilico).	Compresses discovery to 12-18 months, halves preclinical identification costs, and reduces animal testing reliance.	Deep Learning (e.g., AlphaFold), Generative AI, Large Language Models (LLMs), and Quantum Physics Models.
Clinical Development & Trials	Patient matching, synthetic control arms, and "Digital Rehearsal" trial simulation (e.g., Deep Intelligent Pharma).	Halves clinical trial design durations, reduces human enrollment needs, and utilizes multi-agent systems to accurately execute trial workflows.	Natural Language Processing (NLP), Predictive Machine Learning, Generative AI (Digital Twins), and Multi-Agent Systems.
Regulatory Approval	Automated clinical document generation, statistical reasoning, and compliance validation (e.g., Deep Intelligent Pharma).	Cuts documentation time by over 90%, accelerates submissions by 75%, and can achieve zero-revision approvals from regulators.	Large Language Models (LLMs), Natural Language Processing (NLP), Generative AI, and Multi-Agent Systems.
Manufacturing & Commercialization	Process scale-up modeling, real-time quality control, smart manufacturing, and adverse drug reaction monitoring.	Increases throughput by 20-30%, prevents physical batch failures, and reclaims up to 40% of pharmacovigilance capacity.	Computer/Machine Vision, Predictive Machine Learning, and Natural Language Processing (NLP).

Source: Company reports.

Pillar 2: The Unique Chinese Pharma Infrastructure Coupled with AIDD Nurtures Innovations

"Why China?" The true competitive advantage of Chinese AIDD players extends beyond software code; it resides in the physical capacity to rapidly execute and scale. China's unique supply chain density, cross-disciplinary talent pool, and industrialization capacity provide an unmatched infrastructure to translate digital AI predictions into tangible biomedical assets while generating robust proprietary data.

- **We believe Chinese AIDD player like XtalPi and Insilico are structurally better positioned than overseas peers**, thanks largely to China's deep drug development infrastructure. A mature ecosystem of CROs/CDMOs, large clinical trial capacity, and abundant scientific talent lets these platforms run high-throughput experimentation and advance candidates faster and far more cheaply. This helps explain how Insilico's ~12–18 month discovery cycle and 20+ clinical/IND-enabling assets rival Recursion's ~18 month target-to-IND benchmark and seven wholly-owned programs, but at lower cost and greater breadth. Their asset-light, platform-and-services model also generates recurring near-term revenue, whereas Recursion (~\$0.95 EPS 2026E) and Schrödinger remain loss-making due to heavy pipeline spend — making Chinese players more capital-efficient and resilient.

Figure 19: Comparison between China and overseas AIDD players, which are quite different from each other, and deal value signed seems to be higher for China AIDD versus overseas players

Dimension	XtalPi (2228.HK)	Insilico (3698.HK)	Recursion (RXRX)	Schrödinger (SDGR)	Relay Tx (RLAY)
Business model	Service-led AI + robotics platform: discovery services, AAS software, and equity incubation of spin-outs (Signet, METIS); owns wet-lab/robotics	Vertically integrated originator: BD/out-licensing + drug discovery fees + emerging software (MMAI Gym); owns automated labs	Integrated "AI OS" originator + pharma partnerships (Roche, Sanofi); owns large automated wet-lab; minimal external software	Software-licensing core + proprietary pipeline; software-centric; no wet-lab	Pure clinical-stage drug developer; out-licensing (Elevor); no software/wet-lab
Core platform	Quantum-physics + AI + robotics ("Physical AI"); XtalFold Ultra, XenProTOMFNN, Multi-Agent	Pharma.AI PandaOmics, Chemistry42, inClinico, Life Star automated labs	Recursion OS on the "BioHive" supercomputer; phenomics/imaging dataset	Physics-based + AI molecular simulation (FEP+)	Dynamo: motion-based structural biology + AI
Lead clinical asset	Incubated/partnered (SIGX1094)	Renfocarb (Ph II, IPF)	REC-4881 (Ph II, FAP)	SGR-1505 / SGR-3515 (Ph I)	Zovegalisib / RLY-2608 (Ph 3)
Pipeline depth	Largely partner/incubated-owned; few wholly-owned clinical assets	Deep: ~30-asset pipeline, multiple clinical-stage	~5+ clinical + ~15 discovery programs post-2025 cuts	Small proprietary (2–3 clinical) + broad collaborative portfolio	Focused: lead + few follow-on programs
Core partners	Eli Lilly (Ailix, US\$345mm), Pfizer, J&J/Janssen, UCB, DoveTree, Gan Lee	Eli Lilly (US\$2.75bn), Exelixis, Stemline/Menarini, Sanofi	Roche/Genentech, Sanofi, Bayer, Merck KGaA	Numerous (subscription), BMS, Otsuka, Novartis collaborations	Elevor (out-licensing), Genentech (historical)
FY25 total revenue	RMB803mm (~US\$112mm), +20% YoY	US\$56.2mm, ~34.5% YoY	US\$74.7mm	US\$255.9mm, +23.3% YoY	US\$15.4mm
FY25 gross margin	~70% (up from ~46% FY24)	82%	5% (low, partnership mix)	56%	n/m (pre-commercial)
FY25 net profit/(loss)	Profitable: RMB135mm	(US\$352.3mm)	(US\$644.8mm)	(US\$103.3mm)	(US\$276.5mm)
Consensus FY26–28 total sales CAGR	29% per JPMe	74% per JPMe	~40% per bbg consensus	~7% per bbg consensus	~39% per bbg consensus
Consensus FY30 Sales	~US\$50mm	~US\$500mm	~US\$30mm	~US\$375mm	~US\$450mm
Profitability status	First profitable FY25	Turning profitable in FY26	Loss-making in next 3 years	Loss-making in next 3 years	Loss-making in next 3 years
Market Cap (Mid-Jul-2026)	US\$3.9bn	US\$3.2bn	US\$1.6bn	US\$1.1bn	US\$4.2bn
Total out-licensing deal value since 2021 (headline)	~US\$6.9bn: - DoveTree US\$6.0bn - Lilly bispecific US\$345m - Lilly small molecule US\$250m; - excl. undisclosed Pfizer/Janssen/UCB)	~US\$8.0bn+: - Lilly US\$2.75bn - SK US\$2.5bn - Sanofi US\$1.2bn - Exelixis ~US\$0.98bn - Servier US\$888m - Menarini/Stemline ~US\$0.55bn - excl. 2.5bn Bora CDMO alliance)	>US\$20bn: - ~US\$12bn milestone potential across 40 small molecule assets w/ Roche/Genentech - Bayer US\$1.5bn - Sanofi & Merck KGaA via Exscientia	~US\$5.4bn: - Novartis US\$2.3bn - BMS up to US\$2.7bn - Lilly US\$425m etc.	~US\$1.2bn: - Elevor US\$500m - Genentech SHP2 ~US\$695m, terminated Jul 2024
Sum of total out-licensing deal value since 2025 (headline)	~US\$6.4bn	~4.2bn	No deal	~2.3bn	~500mm

Source: Company reports.

- **China's dense R&D infrastructure converts AI predictions into validated assets faster and more cheaply than Western alternatives, anchoring the sector's structural cost advantage.** According to the company, China has compressed the timeline from early discovery to IND application by an estimated 50–70%, while direct clinical execution costs run roughly 30–60% below US and EU levels. This edge is underpinned by a deep network of established investigator sites, a large professional workforce, and a 1.4 billion-person, often treatment-naïve patient pool concentrated near major hospitals, enabling patient recruitment that is 2–5x faster than the West. For AIDD platforms, this matters because the binding constraint is rarely molecule generation—it is the speed and cost of the "make-and-test" validation loop that turns model output into real-world experimental data, a step

where China's clinical and pre-clinical infrastructure provides a durable advantage.

- Regulatory tailwinds are accelerating AIDD execution, as the NMPA shortens review timelines and builds AI-specific frameworks that lower the friction of moving candidates into human testing.** Since China's ICH accession, average review timelines have fallen by roughly 40%, and in October 2025 the NMPA formalized a 30-day IND review-and-approval pathway for eligible innovative drugs—half the prior 60 working days—alongside priority review, breakthrough, and conditional-approval channels, facilitating NDA (new drug application) review. China also permits investigator-initiated trials (IITs) without a formal IND for certain modalities, allowing compounds to enter human testing more quickly than under the FDA framework. In parallel, the NMPA has issued AI-specific guidance and full-lifecycle supervision measures for AI-powered medical products since 2019, creating a clearer regulatory runway for AI-enabled development than many overseas peers. Meanwhile, The FDA's January 2025 AI-in-drug-development guidance and Good AI Practice (GAIP) framework signal that the FDA is actively building pathways for AI-originated data, not blocking them.

Figure 20: Regulatory environment in China continues to turn favorable towards innovation, with AIDD as a key item for action

Date	Jurisdiction	Event	Significance to AIDD
Apr-21	China (CDE, under NMPA)	Guidelines for RWD Used in Generating RWE, supporting drug R&D and review	Legitimized non-traditional data (RWD) for approval, a foundation for AI-driven evidence generation and trial design
May-23	US (FDA, CDER)	CDER discussion paper: "Using AI & ML in the Development of Drug & Biological Products"	First formal FDA signal, opening public dialogue and inviting comment on AI's role across the drug lifecycle
Nov-24	China (NHC/NMPA-linked)	Guidance on AI Application Scenarios in Health & Wellness (84 scenarios)	Provided a concrete roadmap for AI adoption across diagnosis, treatment, and research, supporting demand for AIDD platforms
Jan-25	US (FDA)	Draft guidance: "Considerations for the Use of AI to Support Regulatory Decision-Making for Drugs & Biologics" (7-step credibility framework)	First FDA guidance dedicated to AI in drug development; risk-based model-credibility standard—but explicitly excludes drug discovery itself
Apr-25	China (State Council/NMPA-linked)	Implementation Plan for Digital Transformation of the Pharma Industry (2025–2030)	Set a target of 10+ large-model innovation platforms by 2027, incentivizing AI integration into discovery and development
Jul-25	China (NMPA-linked)	Opinions on Deeply Implementing the "AI+" Initiative	Accelerated AI deployment in healthcare and explicit support for AI-driven drug discovery solutions
Oct-25	China (NMPA)	Research program on standard datasets for AI-powered medical devices	Addressed data quality/standardization—a core bottleneck for training and validating AIDD models
Jan-26	US (FDA) + EMA	"Guiding Principles of Good AI Practice in Drug Development" (10 principles)	First FDA–EMA harmonized principles (human-centric, risk-based, GxP-aligned), signaling international convergence on AIDD standards
Jun-26	China (NMPA)	"Implementation Opinions on AI + Drug Regulation" long-term strategy	Most ambitious NMPA move: integrate AI across the full regulatory lifecycle, targeting an integrated system by 2030 and mature AI ecosystem by 2035

Source: Company reports. Notes: CDE = Center for Drug Evaluation (China); BTD = Breakthrough Therapy Designation; ODD = Orphan Drug Designation; BCC = Biotechnology Company of Concern; FVC = Forced Vital Capacity; NRDL = National Reimbursement Drug List.

- The proprietary data, including real-world clinical data from public hospitals, allow Chinese AIDD companies to leverage high-quality biological, pharmaceutical, and clinical data across many sources. In July 2026, both NDA (National Data Administration) and NHSA (National Healthcare Security Administration) specifically discussed how to trade medical data in the market at their press conference.** If successfully executed, this could allow China to become the first major economic power to allow AI/AIDD models to take advantage of a vast amount of real-world clinical data. XtalPi serves 17 of the top 20 global pharmaceutical companies and has worked with more than 300 biotech and pharma clients, while METiS has built over 30 global partnerships, more than 100,000 wet-lab data points, and a library exceeding 10 million proprietary lipid structures—data

assets that pure-software competitors cannot easily replicate. National policy reinforces this flywheel: China's 15th Five-Year Plan positions AI as a core engine of new productive forces, targeting core digital-economy industries at 12.5% of GDP by 2030 and an AI-related industry exceeding RMB10 trillion, with explicit support for biomedicine. The result is a self-reinforcing loop in which more projects generate more proprietary data, which sharpens vertical AI models, which improves delivery speed and reliability, which in turn attracts more projects. We believe this is an advantage rooted in industrial capacity rather than AIDD software alone.

Figure 21: China's drug R&D infrastructure has cost and time advantages compared to overseas

Dimension	China	US / EU
Discovery-to-IND timeline	Compressed by ~50–70% via parallelized workflows, dense CRO ecosystems, and a culture of executional intensity	Conventional baseline timelines
Clinical trial execution cost	~30–60% below Western levels	Benchmark (higher) cost base
Patient recruitment speed	2–5x faster enrollment (late-stage); large, often treatment-naïve pool	Slower enrollment; more fragmented sites
IND review timeline	30 working days under 2025 NMPA pathway for eligible Class I innovative drugs (from 60)	US FDA standard 30-calendar-day IND review; EU CTR timelines longer (~60 days)
Early human-testing access	ITs allow certain modalities (cell/gene, radioligand, stem cell) into humans without a formal IND; a clearer CGT-IIT pathway is being formalized (effective May 1, 2026) and is gated to credentialed Tier-1 hospitals	Formal IND/CTA generally required first
AI-specific regulation	NMPA full-lifecycle supervision of AI medical devices; formal guidance published 2021–2022 (groundwork from ~2019)	Emerging AI frameworks; less consolidated guidance
National policy support	15th Five-Year Plan; core AI industry targeted to surpass RMB 10tn within the plan period (from ~RMB1.2tn in 2025)	No comparable centralized industrial mandate

Source: Company reports.

Pillar 3: Commercial Validation & Clinical Proof-of-Concept of AIDD

"Why invest in AIDD now?" While the closed-loop systems establish the technological efficiency of AIDD, commercial partnerships and imminent late-stage clinical readouts provide the ultimate financial and scientific validation. By securing lucrative out-licensing agreements and aggressively pushing AI-designed drugs into human trials, these companies effectively shift large capital burdens to Big Pharma while advancing toward the sector's definitive valuation catalysts.

Figure 22: We view the upcoming clinical catalysts for the AIDD space as those with FIC potential

Company	FIC Asset	Target — Indication	FIC Basis	Stage	Key Catalyst & Timing
Insilico (HK 3696)	Rentosetib (ISM001-055)	TN1K — IPF	First TN1K inhibitor in clinic; most advanced AI-discovered/ designed drug globally	Ph III	FPI 2026; topline ~2029
Insilico (HK 3696)	Garutadustat (ISM5411)	Gut-restricted PHD inhibitor — IBD/UC	Potential FIC gut-restricted HIF-PHD inhibitor	Ph IIa	BETHESDA Ph IIa dosing began Jan 2026; topline ~2027
XtalPi (HK 2228) — via Signet	SIGX2649	Pan-TEAD — solid tumors	Potential FIC pan-TEAD inhibitor	IND initiated (Jan 2026)	Preclinical data AACR 2026 (Apr); Ph II entry ~3Q 2026. First clinical readout in ~2027.
Recursion (RXRX)	REC-1245	RBM39 degrader (molecular glue) — solid tumors/lymphoma	Potential FIC RBM39 molecular-glue degrader	Ph 1/2 (DAHLIA)	Prelim ORR readout I n2H26
Relay (RLAY)	Zovogalisib (RLY-2608)	PI3Kα (pan-mutant) — HR+/HER2– BC	First allosteric pan-mutant-selective PI3Kα inhibitor to enter clinical development	Ph III	Ph 3 topline readout in 2027–2028E; FDA Breakthrough Therapy (Feb 2026)

Source: Company data.

- Landmark out-licensing deals with global Big Pharma provide multi-billion-dollar validation, while early-stage asset monetization turns cash-burning biotechs into non-dilutive revenue engines.** Insilico Medicine anchored its platform with a US\$2.75 billion strategic out-licensing collaboration with Eli Lilly (US\$115 million upfront) and an oncology deal with Servier, while XtalPi secured a US\$345 million multi-target large-molecule agreement with Eli Lilly. Beyond platform endorsement, Insilico monetizes individual assets early to fund operations without dilution, licensing its USP1 inhibitor (ISM3091) to Exelixis for US\$80 million upfront, its KAT6 inhibitor (ISM5043) to Stemline for US\$12 million upfront, and a KIF18A inhibitor (ISM9682) to Stemline for US\$20 million upfront against over US\$550 million in total deal value.

Figure 23: Out-licensing activities within the AIDD space have become more diversified, with pharma/biotechs and even CXOs partnering with AIDD companies

Announced	Company	Partner	Scope	Upfront / Near-term	Total Deal Value
Jul-26	Insilico	Bora Pharmaceuticals	Multi-target strategic alliance (AI discovery + dev/mfg)	—	Could exceed ~US\$2.5bn
Jul-26	Insilico	Takeda	Pharma.AI discovery; exclusive worldwide rights	~US\$60m	Up to ~US\$600m + royalties
Jun-26	XtalPi	Undisclosed intl biopharma	Best-in-class small molecule vs GPCR	—	Over ~US\$400m + royalties
Jun-26	Insilico	SK Biopharma	CNS Neuroimmune collab; SK leads late-stage development & commercialization	~US\$18m	Exceed US\$2.5bn + royalties
Mar-26	Insilico	Eli Lilly	Global R&D collab; multiple oral targets	US\$115m	Up to US\$2.75bn + royalties
Jan-26	Insilico	Servier	Oncology R&D collaboration	Up to US\$32m	Up to US\$888m + royalties
Jan-26	Insilico	Qilu Pharmaceutical	Cardiometabolic small-molecule inhibitors	—	Near US\$120m + royalties
Nov-25	XtalPi	Eli Lilly (via Ailux)	AI bispecific antibody + platform license	Double-digit US\$m	Up to US\$345m
Aug-25	XtalPi	DoveTree Medicines	AI+robotics small-molecule & antibody; exclusive global rights	US\$51m + US\$49m near-term (US\$100m)	Up to ~US\$5.99bn + royalties
Jun-25	XtalPi	Pfizer	Next-gen molecular modeling platform (expanded)	—	Undisclosed
Jan-25	XtalPi	UCB (via Ailux)	XtalFold biologics AI platform license	—	Undisclosed
Jan-25	Insilico	Stemline (Menarini)	Solid-tumor asset (ISM9682 / MEN2501)	US\$20m	Over US\$550m + royalties
Nov-24	XtalPi	Janssen / J&J (via Ailux)	Biologics AI platform license	—	Undisclosed
Sep-24	Generate Biomedicines	Novartis	Protein therapeutics discovery	US\$65m	Over US\$1bn + royalties
Feb-24	VantAI	Bristol Myers Squibb	Molecular glue discovery	—	Up to US\$674m
Jan-24	Isomorphic Labs	Eli Lilly	AI small-molecule discovery	US\$45m	~US\$1.75bn
Jan-24	Isomorphic Labs	Novartis	AI small-molecule discovery	US\$37.5m	~US\$1.24bn
Jan-24	Insilico	Stemline (Menarini)	ISM5043 (KAT6 / MEN2312) out-license	US\$12m	Over US\$500m + royalties
Sep-23	Insilico	Exelixis	ISM3091 (USP1) out-license	US\$80m	~US\$955m + royalties
May-23	XtalPi	Eli Lilly	AI small-molecule drug discovery	Undisclosed	Up to US\$250m

Source: Company data.

Figure 24: ... And 2026 total upfront and headline deal value related to AIDD has reached a new high, indicating strong market potential

Year	Total Upfront (US\$m)	Total Headline Deal Value (US\$bn)
2023	80	1.21
2024	159.5	5.16
2025	130	6.89
2026 (YTD)	225	9.76

Source: Company data.

- **By commercializing their proprietary software and testing environments, AIDD platforms are establishing predictable, SaaS-like revenue streams that complement their clinical development pipelines.** Recognizing the broader commercial value of their underlying technology, these companies are directly monetizing their computational architecture. For example, Insilico Medicine launched "Science MMAI Gym," a first-of-its-kind, domain-specific training environment that effectively teaches general-purpose foundation models how to perform complex, real-world drug discovery tasks. By offering this specialized environment through flexible annual subscriptions and pay-per-token cloud licensing, Insilico helps partners boost the chemical and biological accuracy of their AI models by up to ten times, creating a highly scalable, recurring revenue stream. Similarly, XtalPi leverages its "picks and shovels" infrastructure to generate predictable subscription- and transaction-based service fees directly from its software tools and automated chemical synthesis services. Instead of merely relying on long-term clinical milestones, XtalPi monetizes its vast array of predictive software modules and robotic wet labs—often deploying a full-time equivalent (FTE) subscription model—to provide continuous, high-margin R&D support for pharmaceutical clients.
- **Realizing the great potential of AIDD, a number of top AI and technology companies are starting to get into AIDD in various ways recently. Although we are waiting to see whether those companies can be successful in the field of AIDD, the wave of those companies starting AIDD related projects suggests the great potential of AIDD.** In addition to directly partnering with large pharma, those AI and tech companies have developed: 1) life-science-specialized models / "scientist workbenches", and 2) platform + infrastructure plays that make model-driven design and lab feedback loops easier to run. For example, OpenAI launched "GPT-Rosalind," positioned as a life sciences research model for hypothesis generation and analysis. Anthropic launched Claude Science, a workbench integrating many connectors/tools and compute backends for workflows spanning genomics to cheminformatics. Anthropic also acquired Coefficient Bio, a biotech automation platform. One of Google's key footprints in AIDD is its Isomorphic Labs, which has raised significant funding to scale its AI drug design engine and expand pharma partnerships. AWS launched Amazon Bio Discovery, an AI platform for early-stage drug discovery with foundation models + an agent to help configure workflows, explicitly supporting "lab-in-the-loop" iteration. NVIDIA is pushing AIDD via the BioNeMo Agent Toolkit. ByteDance established Anew Labs (Anew Therapeutics) as its internal AIDD unit in 2021; Anew Labs has since matured into an independent spin-off.

We Answer Yes to The Top 3 Critical AIDD Questions

Q1: "Zero Approved Drugs — Is the Clinical Promise of AIDD Real?"

- **The crux of this question is that, despite years of marketing and hundreds of preclinical candidates, no AI-discovered drug has yet received commercial approval in the world, raising the question of whether AIDD companies are selling infrastructure rather than therapeutic outcomes.** As of mid-2026, the global AIDD sector has generated 173 clinical-stage programs, of which 94 are in Phase 1, 56 in Phase 2, and only a handful in Phase 3. However, none have crossed the regulatory finish line. The AI-acceleration thesis has been demonstrated convincingly at early stages (target-to-PCC in 12–18 months versus the industry average of 4.5 years), but the critical question, whether AI-optimised molecules outperform traditionally designed ones in late-stage clinical settings, remains unanswered currently. However, based on the following arguments, we are confident that AIDD will generate drugs which will achieve Ph3 success and gain marketing approval.
- **The counter-argument is strongest at Insilico, where rentosertib (ISM001-055) for IPF has moved from AI-generated drug target and drug molecule to a Ph3 trial with its discovery to clinic path published in *Nature Biotechnology* and Phase 2a data published in *Nature Medicine*, representing a genuine scientific first.** At 60 mg QD, the drug achieved +98.4 mL FVC (forced vital capacity) at week 12 versus –20.3 mL in the placebo arm, earning China CDE Breakthrough Therapy Designation in May 2025. A Phase 3 trial (GENESIS-IPF-3, n=192, 52-week primary endpoint of FVC change) has been initiated in China in 2026, with topline data expected by 2029, while a US Phase IIB is simultaneously underway.
- **Other examples of promising drugs discovered or optimized through AIDD** include Recursion's REC-1245, which is a highly potent, first-in-class RBM39 (RNA-binding motif protein 3) degrader in Ph1, where RBM39 is a vital pre-mRNA splicing factor that regulates gene expression and cell survival. Generate Biomedicines' GB-0895 is the first long-acting TSLP mAb in Ph3 development. According to the company, it has the best profile among all known TSLP targeting programs. Schrödinger and Nimbus discovered the TYK2 inhibitor TAK-279, which was acquired by Takeda in 2023 and is currently being evaluated in Ph3 for psoriasis.
- **As AI helps solve or provide new solutions to numerous long-standing scientific problems in physics, chemistry, and mathematics, which have puzzled humans for years, we believe AI will be able to eventually discover drugs that can achieve global Ph3 trial success.** We think LLMs are quickly improving their frontier R&D ability. For example, for the OpenAI FrontierScience-Research Benchmark, the best model at that time, GPT-5.2, achieved only 25% accuracy in 2025. However, with the release of GPT-5.6, we see its accuracy in various science benchmarks is consistently ~20% (absolute number) higher than GPT-5.4. Hence, we think GPT-5.6 might achieve 50% in the FrontierScience-Research Benchmark. In specific subjects, we see AI models help solve or improve answers to long-standing problems. In physics, AI models have advanced our knowledge in the 150-

year fluid dynamics dilemma in Navier-Stokes equations, loop geometries in gravitational radiation in string theory, and the electron spin problem in magnets in quantum physics. In chemistry, AI helps generate 2.2 million potential new crystals, 380,000 of which could be stable; AI partially solves the Schrödinger Equation to understand electron interaction, and designs chemical synthetic pathways for retrosynthesis. In math, AI discovered a new family of solutions for the Erdos Unit Distance problem and the Cap Set problem. And AI broke long-standing records for five complex Ramsey numbers. If AI models already have strong capabilities in solving difficult scientific problems, agents and self-improving features can only make AI models more powerful in solving complex AIDD problems.

- **Our view is constructive: AI has been proven mainly in language and code, but the same first-principles rigour is now extending into the life sciences.** Biology is harder — human systems are only partially understood, cell-line and animal results don't always translate to patients, and DMPK and toxicology remain real bottlenecks; even AI-native programs still face the better part of a decade to approval, and no fully AI-designed drug has yet been approved. Still, the trajectory is encouraging: in AI-driven materials science, physics-based modelling and autonomous "self-driving" labs aim to compress discovery from 10–20 years to just 1–2 — screening battery-electrolyte chemistries down to a tested shortlist via active learning and accelerating perovskite solar-cell design through robotic synthesis. Since biology obeys the same physical laws that XtalPi's quantum-physics engine exploits, our thesis is that as first-principles data compounds through wet-lab flywheels, the translational gap should narrow — though AI-discovered molecules still revert toward the historical ~40% success rate in Phase II, so early proof points must survive the clinic to become approvals.

Q2: "Does China Origin Imply Structural Discount — Can AIDD Drugs and Platforms Go Global?"

- **China-domiciled AIDD companies might be facing a so-called "China discount" driven by (i) US legislation and administrative orders aimed at curbing China's biotech sector, which threaten future data-sharing and licensing/collaboration deals, (ii) FDA/NMPA regulatory divergence that complicates multi-regional trial strategies, and (iii) the NRDL (national reimbursement drug list) price cuts could limit the upside of commercialisation economics in China.** In June 2026, Representatives John Moolenaar and Debbie Dingell introduced the bill of Biotech Investment National Security Act (BINSa). This legislation would amend the COINS framework to subject U.S. pharmaceutical licensing agreements, joint ventures, and equity investments in Chinese biotech entities to strict Treasury and Defense Department reviews. In addition, the COINS Act already gives the Treasury Department the discretionary authority to expand its "covered sectors" to include biotechnology through the rulemaking process without needing additional legislation. The House Appropriations Committee advanced an Agriculture/FDA spending bill report (spearheaded by Rep. Andy Harris) seeking to bar the FDA from accepting or reviewing clinical trial data generated in China in support of IND (investigational new drug) applications. However, we don't expect a material impact from those actions in the near-term.
- **The regulatory convergence argument that US-China AIDD collaboration is ultimately in both parties' interest is gaining traction among scientists even as politicians diverge.** Published analysis notes that indiscriminate restrictions on collaboration risk undermining partnerships that are accelerating therapeutic

progress, and that intelligence concerns about IP transfer apply to specific firms rather than the sector wholesale. The FDA's Good AI Practice guidelines and NMPA's parallel 2026 AI roadmap suggest a pragmatic regulatory evolution toward accepting AI-generated evidence from credible, well-documented sponsors, regardless of country of origin.

- **A pointed investor question worth addressing directly is whether the systematic exclusion of Chinese companies and users from leading US frontier AI models, most visibly Anthropic's September 2025 ban on Claude access for all China-controlled entities, structurally disadvantages Chinese AIDD platforms relative to US peers.** We don't think so given China-originated AI LLMs continue to show on-par capabilities with their US peers. Anthropic announced on 5 September 2025 that it would prohibit any company more than 50% owned by a Chinese entity from accessing Claude's API, citing national security and misuse risks; this followed the pre-existing blocks on ChatGPT, Gemini, and Meta AI in mainland China. The GPU chip export controls reinforced the access asymmetry: US Commerce BIS regulations progressively restrict advanced semiconductor shipments to China, with the May 2025 BIS guidance explicitly extending controls to Chinese-headquartered firms' overseas subsidiaries. The result is a structural compute and model-access gap: as of mid-2026, the best Chinese frontier models (DeepSeek V4) lag US frontier models by approximately 6–7 months on general benchmarks, and DeepSeek's own technical paper concedes it "trails state-of-the-art frontier models by approximately 3 to 6 months". However, on July 16, 2026, Chinese company Moonshot released Kimi K3, a highly disruptive LLM. Packing an estimated 2.8 trillion parameters and a 1M token context window, it achieves benchmark parity with frontier US models like OpenAI's GPT-5.6 Sol and Anthropic's Claude Fable 5, while undercutting them significantly in price.
- **Hence, the China discount argument towards AIDD is more nuanced than the general AI narrative suggests.** First, AIDD platforms like Pharma.AI and XtalPi's quantum physics engine are domain-specific systems built on proprietary biological and chemical datasets, physics-based algorithms, and wet-lab feedback loops — their core value is not derived from access to GPT or Claude, but from in-house first-principles scientific modelling and experimental validation. Second, Chinese models have shown surprising competitiveness in the specific scientific subdomains most relevant to drug discovery: DeepSeek's coding and mathematical reasoning benchmark scores are broadly comparable to US peers, and Insilico's own MMAI Gym training data showed a Qwen3 model outperforming all frontier foundation models on clinical target retrieval benchmarks after a single training session. Third, the hardware constraint has historically driven Chinese AI efficiency innovation rather than absolute capability collapse — DeepSeek R1's efficiency at a fraction of US compute costs is the most prominent example. Kimi K3 token cost is also much lower than US frontier models.
- **Our conclusion is that the compute access gap is a legitimate risk that might warrant further consideration of valuation for general-purpose Chinese AI companies, but for domain-specific AIDD platforms whose competitive moat lies in proprietary scientific data, specialized AI model training, and physical wet-lab infrastructure, the discount is materially smaller than the headline geopolitical narrative implies.** The more important performance arbiters for AIDD are clinical trial outcomes and BD deal quality — domains where both Insilico and XtalPi are demonstrably competitive on a global stage.

Q3: "Revenue Model can be Lumpy and Unsustainable — Is AIDD a Real Business?"

- **One critique of China AIDD companies is that their revenue streams are often driven by one-time, large-ticket BD events that are inherently unpredictable and create year-on-year volatility, making financial forecasting unreliable for institutional investors.** Insilico's FY2025 results illustrate this case: total revenue fell 34.5% YoY from US\$85.8M to US\$56.2M, almost entirely because pipeline development revenue dropped from US\$76.6M to US\$23.9M — a 68.8% collapse — as the Lilly US\$115M upfront slipped into 2026. Management itself disclosed the business "operates under a project-based business model with no guarantee or clear visibility on future revenue generation". XtalPi's trajectory is the positive mirror image: FY2025 revenue surged 201% to RMB802.6M, but Customer A (undisclosed) accounted for ~RMB365M — approximately 45% of total revenue — creating some concentration risk.
- **AIDD companies are attempting to address this structural fragility through recurring-revenue layers, but the transition is still early-stage.** Insilico is building its Software Solutions subscription business (Pharma.AI platform: Biology42, Chemistry42, Medicine42, Science42), which grew 23.8% to US\$4.9M in 2025 as the subscriber base expanded from 153 to 181 customers. More ambitiously, the Science MMAI Gym — a membership-based AI training environment for frontier foundation models — aims to generate recurring licence fees; its first partnership with Liquid AI was inked shortly after launch. XtalPi's AI4S Intelligent Solutions segment (RMB264.7M, up 62.6% in FY2025) represents platform deployments for pharmaceutical and materials-science clients, including 17 of the world's top 20 pharmaceutical companies.
- **The key investor question is whether BD deal value can be recycled into a more annuity-like model to support long-term cash runway.** XtalPi ended FY2025 with RMB7.1B in total cash and raised a further RMB2.5B in convertible bonds in early 2026. Insilico ended FY2025 with US\$393M in cash post-IPO, with the Lilly US\$115M upfront in Q1 2026 meaningfully extending cash visibility. The bear case remains that until recurring software/platform revenues constitute >30–40% of total revenues (currently ~10–15% for both companies), AIDD financials will remain BD-event-driven, which carries some uncertainty in timing.
- **Crucially, however, the lumpiness of recognised revenue should not be confused with a slowdown in underlying deal formation — BD momentum at both companies is, by any measure, accelerating.** For Insilico, the post-IPO deal streak has been exceptional: the Lilly global R&D collaboration (US\$2.75B total deal value, US\$115M upfront, March 2026), a Servier oncology deal (up to US\$888M, 2025), a Tenacia CNS expansion (up to US\$94.75M, March 2026), and most recently a record-breaking SK Biopharmaceuticals CNS partnership (total deal value exceeding US\$2.5B, June 2026) — making it the company's largest-ever APAC partnership — mean Insilico has contracted well over US\$7.5B in aggregate potential deal value since 2021. For XtalPi, the Eli Lilly relationship has deepened into a second landmark: subsidiary Ailux sealed a US\$345M AI-powered bispecific antibody partnership in November 2025, making XtalPi one of the very few AIDD companies globally to have major collaborations with a top-5 pharma in both small molecule and large molecule discovery. The cumulative deal pipeline and the quality of counterparties (Lilly, Servier, SK Biopharm, Stemline, Qilu) demonstrate that the largest global pharmaceutical companies continue to allocate meaningful capital to

China AIDD platforms — validating the underlying technology. We think with continued new BD deals and milestone payments emerging, AIDD companies' revenue could become more even.

Investment Risks

- **Clinical Failure Risk.** Rentosertib is the most advanced pure-play AI-discovered drug globally, making its upcoming Phase III readout a sector-defining event. Despite strong Phase IIa efficacy and regulatory recognition, IPF remains a difficult indication with historically low approval success rates. A Phase III failure would likely trigger a broad de-rating across the AIDD sector, given no AI-discovered drug has yet achieved global commercial approval.
- **Lumpy, Licensing-Driven Revenue Model.** Insilico and XtalPi remain heavily dependent on milestone-based licensing and business development (BD) deals, resulting in highly volatile and difficult-to-forecast revenues. Insilico's FY2025 revenue decline reflected timing shifts in major licensing payments, while XtalPi remains reliant on a small number of large customers. With recurring platform revenues still limited, financial performance remains largely driven by the timing and size of partnership transactions.
- **COINS Act & US Cross-Border Licensing Risk.** The proposed BINS Act and passed COINS Act highlight growing US scrutiny of collaborations involving Chinese biotech and AI companies, creating regulatory and political uncertainty around future licensing, co-development, and commercialization agreements with US partners. Even without direct restrictions, heightened compliance requirements and geopolitical concerns could slow deal-making, reduce partner appetite, and increase execution risk for China-based AIDD firms seeking access to US markets and capital.
- **FDA Regulatory Uncertainty for AI-Generated Evidence.** Emerging FDA guidance on AI in drug development could require greater disclosure of model architecture, training data, and validation methods for high-impact AI systems. For companies whose competitive advantage rests on proprietary AI platforms, balancing regulatory transparency with IP protection remains an unresolved challenge, potentially complicating future regulatory submissions.
- **AIDD Model generalization risk.** A May 2026 Nature Structural & Molecular Biology study (Škrinjar et al.) exposed a key weakness in the AI tools that predict how a drug molecule fits into a protein—the core task behind AI drug discovery. The researchers tested leading models (AlphaFold3, Chai-1, Boltz-1, Protenix) on 2,600 protein–ligand structures the models had never seen during training. On familiar cases, accuracy was about 70%; on genuinely new ones, it collapsed to single digits (AlphaFold3 ~9%, Boltz-1 ~5%). In plain terms, the AI was largely copying answers it had memorized rather than truly figuring out where new molecules bind—and it couldn't tell when it was wrong, since its confidence scores had almost no link to actual accuracy. Adding more training data (Boltz-2 vs. Boltz-1) did not help. Because drug discovery is fundamentally about brand-new, uncharted targets, this challenges the “structure prediction is solved” narrative and favors platforms that pair AI with physics-based validation to double-check predictions.
- **Unproven Clinical Advantage of AI Discovery.** The core AIDD thesis assumes AI can improve drug development productivity and success rates, yet there is still no

clinical evidence that AI-discovered molecules outperform conventionally discovered drugs in late-stage development. Until multiple AI-originated programs achieve Phase III success and regulatory approval, the industry's ability to translate preclinical efficiency gains into superior clinical outcomes remains unproven.

Figure 25: Abbreviations

Abbreviation
AIDD — AI-driven drug discovery.
AI — Artificial intelligence.
R&D — Research and development.
CRO — Contract research organization.
CDMO — Contract development and manufacturing organization.
IND — Investigational new drug application.
NMPA — National Medical Products Administration, China's drug regulator.
FDA — US Food and Drug Administration.
CDE — Center for Drug Evaluation in China.
CAGR — Compound annual growth rate.
BD — Business development, typically referring to collaborations, licensing, and partnerships.
SOTP — Sum-of-the-parts valuation.
rNPV — Risk-adjusted net present value.
DCF — Discounted cash flow valuation.
EV/S — Enterprise value to sales multiple.
P/S — Price to sales multiple.
PoS — Probability of success.
WACC — Weighted average cost of capital.
PCC — Pre-clinical candidate.
SaaS — Software as a service.
DMTA — Design-make-test-analyze workflow.
IPF — Idiopathic pulmonary fibrosis.
FVC — Forced vital capacity, a lung-function measure.
TNIK — Traf2- and Nck-interacting kinase, the target of Rentosertib.
ENPP1 — Ectonucleotide pyrophosphatase/phosphodiesterase 1, a target in immuno-oncology.
STING — Stimulator of interferon genes pathway, involved in immune activation.
ADMET — Absorption, distribution, metabolism, excretion, and toxicity.
PKPD — Pharmacokinetics and pharmacodynamics.
LLM — Large language model.
API — Application programming interface.
GPU — Graphics processing unit, used for AI computation.
IIT — Investigator-initiated trial.
GAIP — Good AI Practice framework.
BTB — Breakthrough Therapy Designation.
ODD — Orphan Drug Designation.
BCC — Biotechnology Company of Concern.
NRDL — National Reimbursement Drug List in China.
CAR-T — Chimeric antigen receptor T-cell therapy.
LNP — Lipid nanoparticle.
mRNA — Messenger RNA.
PRMT5 — Protein arginine methyltransferase 5, an oncology target.
GLP-1R — Glucagon-like peptide-1 receptor.
IBD — Inflammatory bowel disease.
USAN — United States Adopted Name.
AUC-ROC — Area under the receiver operating characteristic curve, a predictive model accuracy metric.
FTE — Full-time equivalent, often used as a services contract model.
YoY — Year over year.
APAC — Asia-Pacific.
CNS — Central nervous system.
IP — Intellectual property.
AI4S — AI for Science solutions.

Initiation

Overweight

2228.HK, 2228 HK

Price (23 Jul 26):HK\$7.04

Price Target (Dec-27):HK\$10.00

Healthcare

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XtalPi - H

Service-Led AIDD Platform Differentiated with Diverse Businesses and Robotics; Initiate at OW

We initiate coverage on XtalPi with an Overweight rating, viewing the company as a service-led, diversified way to gain exposure to China's AIDD theme. While sharing the same secular AI-driven R&D opportunity as Insilico Medicine, XtalPi differentiates itself through a diverse set of revenue-generating businesses built on proprietary AI, robotics and laboratory automation infrastructure. The company already monetizes its technology through discovery services and AI4S contracts, supported by partnerships with Lilly, UCB, Janssen and Pfizer and a growing backlog. Beyond services, XtalPi's incubation strategy adds equity-linked upside through portfolio/partner companies such as Signet Therapeutics and METiS TechBio. Although BD related revenue remains inherently lumpy, we believe its combination of commercialized services, strategic partnerships and multiple monetization pathways supports an attractive long-term growth outlook.

- **A Robotics-Enabled Platform Is XtalPi's Core Competitive Advantage.** XtalPi's key differentiator is its integration of AI, computational science and laboratory automation into a closed-loop discovery platform. Rather than operating as a software-only provider, the company combines quantum-physics-based computation, proprietary AI models, multi-agent systems and robotic "Physical AI" capabilities to automate the full design-make-test-analyze (DMTA) workflow. This integration enables faster iteration cycles, scalable experimental execution and the continuous generation of proprietary datasets. The robotics infrastructure currently focuses on small-molecule experiments. However, we believe it is being extended to biologics engineering, while also extending beyond pharmaceuticals into adjacent verticals including materials science and consumer health.
- **A Diverse Set of Businesses Provides Revenue Visibility.** Unlike many AIDD peers that remain dependent on future pipeline monetization, XtalPi already generates meaningful commercial revenue from its platform. The company provides drug-discovery services and AI4S solutions across a diverse customer base, delivering 2025 revenue of RMB803mn, up 41% YoY, while building a RMB7.07bn backlog that supports future growth and a path toward sustained profitability. Commercial validation is evidenced by a partner roster that includes Lilly, Janssen, UCB and Pfizer, with the US\$345mn Lilly/Ailux transaction representing a notable proof point of platform value. That said, revenue recognition remains linked to project milestones and collaboration progress, resulting in periodic lumpiness at the quarterly level.
- **Incubation Creates a Monetization Channel while Capturing Potential Equity Upside.** Beyond services and proprietary pipeline assets, XtalPi has established an incubation model that creates a third, equity-linked growth vector. The company selectively co-founds and supports independent biotech ventures, contributing its AI and robotics platform while retaining ownership stakes. This structure enables XtalPi to participate in the long-term value creation of multiple external programs without bearing the full capital requirements associated with late-stage clinical development. Early examples include Signet Therapeutics, which has advanced SIGX1094 and additional oncology assets, and METiS TechBio, an AI-enabled drug-development platform company. As a result, XtalPi can monetize a single technology platform through service fees, partnership economics and equity appreciation from successful portfolio companies.
- **A visible catalyst path supports medium-term value realization.** XtalPi's

2026–2027 outlook is underpinned by multiple catalysts, including milestone recognition from DoveTree, clinical progression of assets such as SIGX1094 and pan-TEAD SIGX2649, and development updates from partners (e.g., PharmaEngine’s PRMT5 inhibitor PEP08). Ongoing BD activities add further optionality through incremental upfront and milestone payments, with minimal incremental cost given the scalable nature of its infrastructure. Importantly, these catalysts build on an existing revenue base, rather than serving as binary validation events.

- **Valuation & Risk:** Our Dec-2027 PT of HK\$10 is based on a Sum-Of-the-Parts methodology. Key downside risks include unfavorable sentiment towards AI themes, sluggish outlicensing deals with lower-than-expected value, unfavorable clinical trial outcomes, geopolitical risks including rising US-China tension, and worse-than-expected sales & earnings.

Price Performance



Company Data

Shares O/S (mn)	4.303
52-week range (HK\$)	15.12-5.67
Market cap (\$ mn)	3,864
Exchange rate	7.84
Free float (%)	85.9%
3M ADV (mn)	61.45
3M ADV (\$ mn)	61.5
Volatility (90 Day)	60
Index	HSCI
BBG ANR (Buy Hold Sell)	15 0 0

Key Metrics (FYE Dec)

Rmb in millions	FY25A	FY26E	FY27E	FY28E
Financial Estimates				
Revenue	803	1,065	1,446	2,010
Adj. EBITDA	171	(168)	(82)	125
Adj. EBIT	55	(257)	(174)	19
Adj. net income	247	(40)	45	242
Adj. EPS	0.06	(0.01)	0.01	0.06
BBG EPS	(0.03)	(0.01)	0.03	0.08
Cashflow from operations	(165)	(77)	79	264
FCFF	(335)	(265)	(133)	33
Margins and Growth				
Revenue Growth Y/Y (%)	201.2%	32.7%	35.8%	39.0%
EBITDA margin	21.3%	(15.8%)	(5.7%)	6.2%
EBITDA Growth Y/Y (%)	(131.4%)	(198.6%)	(51.5%)	(252.6%)
EBIT margin	6.9%	(24.1%)	(12.1%)	1.0%
Net margin	30.8%	(3.8%)	3.1%	12.0%
Adj. EPS growth	(153.9%)	(116.4%)	(211.4%)	436.2%
Ratios				
Adj. tax rate	0.6%	0.0%	0.0%	5.0%
Interest cover	NM	1.6	0.8	NM
Net debt/Equity	NM	NM	NM	NM
Net debt/EBITDA	NM	23.6	48.2	NM
ROCE	0.8%	(2.6%)	(1.8%)	0.2%
ROE	3.7%	(0.4%)	0.5%	2.5%
Valuation				
FCFF yield	(1.3%)	(1.0%)	(0.5%)	0.1%
Dividend yield	-	-	-	-
EV/Revenue	27.2	20.7	15.3	10.9
EV/EBITDA	128.0	NM	NM	176.2
Adj. P/E	105.8	NM	580.9	108.3

Summary Investment Thesis and Valuation

Investment Thesis

We rate XtalPi at Overweight as a differentiated, service-led way to gain exposure to China's AI drug discovery theme. Unlike pipeline-centric peers that rely heavily on binary clinical outcomes, XtalPi generates revenue today through AI-enabled discovery services and AI-for-Science contracts, leveraging its unique robotics-driven "Physical AI" platform that integrates AI models with automated wet-lab experimentation in a closed-loop discovery engine. This scalable infrastructure creates a durable competitive moat, reinforced by proprietary experimental data and applicability beyond pharmaceuticals into materials science and consumer health. Supported by a diversified roster of blue-chip partners, including Eli Lilly, Janssen, UCB and Pfizer, and a sizable backlog, XtalPi offers a broader and lower-risk commercialization pathway, although revenue remains inherently milestone-driven and uneven. Looking ahead, we see multiple catalysts from partner milestones, asset progression and new business development opportunities, providing attractive upside while building on an already established and growing revenue base rather than depending on binary proof-of-concept events.

Valuation

Our Dec-2027 price target is HK\$10, which is based on SOTP. We break down XtalPi's business into 1) intelligence solutions, 2) drug discovery, and 3) self-developed pipeline, with the first two being valued using an assigned AI-empowered EV/S and the third business at (risk-adjusted net present value) rNPV. The implied 27E P/S is ~25x, which is at the lower end of global AIDD peers trading at 5x-173x (median ~60x).

A Diverse Set of Growth Engines: Proprietary Drug Pipeline, Robotics Services, Pharma Partnerships, AI4S Infrastructure, and Incubation Model

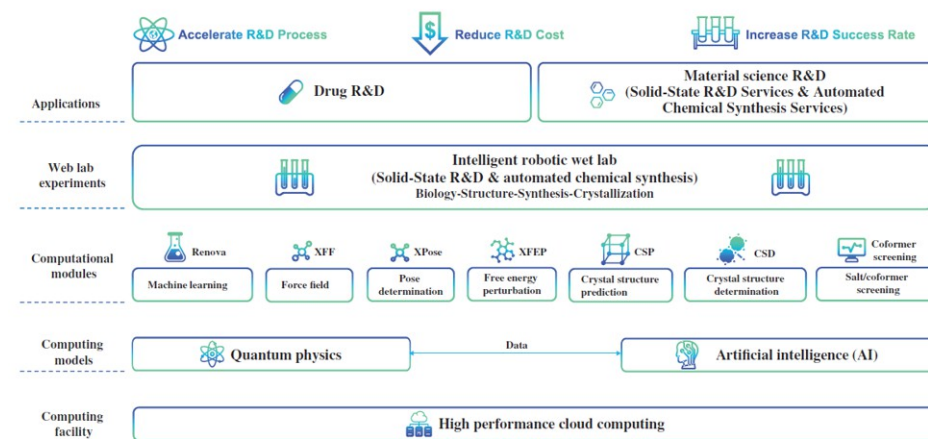
We forecast XTalPi to run three of its own pipeline instead of outlicensing to other pharma. We expect the peak sales for each to reach Rmb2-3bn, making them the primary long-term growth driver for the company. That said, we also expect its AI4S automation business to generate ~Rmb3.5bn sales in 2035E, making the company a more diversified play.

Figure 26: XTalPi's Sales Build

Sales Build	FY2024	FY2025	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E
TOTAL REVENUE (RMB m)	266.4	802.6	1,065.1	1,446.4	2,010.1	2,474.0	3,015.0	3,759.4	5,236.0	6,883.0	8,837.9	11,244.3
Drug Discovery Solutions — segment revenue	103.7	537.9	599.1	777.5	1,110.4	1,298.9	1,553.0	1,966.0	3,076.5	4,302.5	5,780.8	7,656.4
- Self-pipeline (full rights - Risk-adjusted)	-	-	-	-	-	-	-	-	553.3	1,381.5	2,143.8	3,209.1
BCMAxCD3 TCE	-	-	-	-	-	-	-	-	89.2	172.8	229.1	269.6
IL-23/TL1A Bispecific	-	-	-	-	-	-	-	-	159.6	447.8	754.6	1,233.5
Long-acting FcRn	-	-	-	-	-	-	-	-	304.5	781.0	1,160.1	1,706.0
- Collaboration sales (Risk-adjusted)	-	411.2	447.0	595.0	895.0	1,051.2	1,268.2	1,647.0	2,172.3	2,535.0	3,212.4	3,980.2
PoS adjustment	-	100%	100%	100%	85%	70%	60%	60%	60%	60%	60%	60%
- Collaboration sales	-	411.2	397.0	495.0	856.0	1,169.2	1,570.2	2,125.0	2,826.3	3,411.0	4,398.4	5,336.2
DoveTree (RMB Mn)	0	361.2	280.0	350.0	600.0	900.0	1,200.0	1,600.0	2,000.0	2,500.0	3,200.0	3,800.0
Eli Lilly (RMB Mn)	-	50.0	85.0	70.0	120.0	130.0	140.0	290.0	440.0	590.0	740.0	890.0
Gan & Lee Pharma (RMB Mn)	-	-	20.0	15.0	20.0	30.0	40.0	55.0	70.0	100.0	150.0	200.0
Disclosed pipeline drugs	-	-	12.0	60.0	116.0	109.2	190.2	180.0	316.3	221.0	308.4	445.2
- Subscriptions/others	104	127	152.1	182.5	215.4	247.7	284.8	319.0	350.9	386.0	424.6	467.1
AI4S / Intelligent Automation — segment revenue	162.8	264.7	466.0	668.9	899.7	1,175.1	1,462.0	1,793.3	2,159.4	2,580.5	3,057.1	3,588.0
- Robotics	-	69.0	159	254	382	553	747	971	1,214	1,493	1,807	2,150
- Chemical synthesis & others	0.0%	205	307	414	518	622	715	822	946	1,087	1,251	1,438
YoY												
TOTAL REVENUE (RMB m)	52.8%	201.2%	32.7%	35.8%	39.0%	23.1%	21.9%	24.7%	39.3%	31.5%	28.4%	27.2%
Drug Discovery Solutions — segment revenue	18.2%	418.9%	11.4%	29.8%	42.8%	17.0%	19.6%	26.6%	56.5%	39.8%	34.4%	32.4%
- Self-pipeline (full rights - Risk-adjusted)	-	-	-	-	-	-	-	-	#DIV/0!	149.7%	55.2%	49.7%
- Collaboration sales	-	-	8.7%	33.1%	50.4%	17.5%	20.6%	29.9%	31.9%	16.7%	26.7%	23.9%
- Subscriptions/others	18.2%	22.3%	20.0%	20.0%	18.0%	15.0%	15.0%	12.0%	10.0%	10.0%	10.0%	10.0%
AI4S / Intelligent Automation — segment revenue	87.8%	62.8%	76.1%	43.5%	34.5%	30.8%	24.4%	22.7%	28.4%	19.5%	18.5%	17.4%
- Robotics	0.0%	0.0%	165.0%	60.0%	50.0%	45.0%	35.0%	30.0%	25.0%	23.0%	21.0%	19.0%
- Chemical synthesis & others	0.0%	0.0%	50.0%	35.0%	25.0%	20.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%

Source: J.P. Morgan estimates, Company data.

Figure 27: XTalPi's integrated AI platform spans various industries, not just drugs



Source: Company reports.

XTalPi's partnerships are not limited to pharma but also energy companies, and we expect more collabs with players from other industries like material science and agricultural science.

Figure 28: XTalPi's partnerships span different industries

Date	Partner	Asset/Target	Type	Upfront	Total Value
2016–2017	Pfizer	Crystal structure prediction / polymorph screening	Service collaboration	Undisclosed	Undisclosed
May-18	Pfizer	AI/physics molecular-modeling platform (XFF, XFEP)	10-yr strategic Master Collaboration & License	Undisclosed	Undisclosed
Oct-22	Janssen (J&J)	Validated small-molecule hits for nominated target (ID4)	Research collaboration	Undisclosed	Undisclosed
2023	Eli Lilly	Small-molecule	AI + robotics drug discovery collab	Undisclosed	Up to US\$250m
Nov-24	Sinar Mas	Strategic APAC AI	Strategic collaboration	Undisclosed	Undisclosed
Dec-24	NCCS/Duke-NUS	Peptide drug discovery	Academic research	Undisclosed	Undisclosed
Jan-25	UCB	XtalFold biologics	Non-exclusive license	Undisclosed	Undisclosed
Jun-25	Pfizer (expansion)	Next-gen molecular-simulation platform	Strategic extension	Undisclosed	Undisclosed
Jun-25	Tasly	PEP08 (PRMT5)	Incubated/partnered	n/a	Milestone on IND
Jul-25	JingJiaHang	Radionuclide conjugates	Co-development	Undisclosed	Undisclosed
Aug-25	DoveTree	Oncology/autoimmune	Strategic platform	US\$51m+	Up to US\$5.99bn
Sep-25	UAE Royal Office	Traditional medicine	Lab infrastructure	Undisclosed	~US\$30m
Nov-25	Eli Lilly/Ailux	Bispecific antibody	Multi-target strategic	Undisclosed	US\$345m
Dec-25	Gan & Lee	PepiX peptides	Global strategic	Platform fee	Milestone/royalty
Jan-26	JinkoSolar	Solar cells	Materials JV	Undisclosed	Commercial share
Feb-26	VISEN	Endocrine/metabolic	Research collaboration	Undisclosed	Tens of millions
Jun-26	Undisclosed	Oral GPCR small-molecule	Discovery/out-license	Undisclosed	>US\$400m

Source: Company reports.

XtalPi has evolved from providing one-off services to establishing deep platform licensing and co-development partnerships with both global and Chinese collaborators. In biologics, its Ailux subsidiary signed a US\$345mn bispecific antibody agreement with Eli Lilly and licensed its XtalFold platform to Janssen and UCB. In small molecules, XtalPi entered into a multi-billion-dollar pipeline licensing LOI with DoveTree, a company founded by Professor Gregory Verdine. Meanwhile, its PepiX peptide platform underpins a global collaboration with Gan Lee Pharmaceuticals. The company has also expanded its molecular simulation partnership with Pfizer and, in materials science, formed a strategic collaboration with JinkoSolar. According to management, the cumulative value of these partnerships has reached the tens of billions of RMB.

XtalPi's robotic labs are the "Physical AI" execution engine that closes the loop between AI prediction and real-world experimentation, transforming the platform into a self-improving discovery system. Acting as the "precision hands" to AI models' "expert brain," the labs conduct 24/7 high-throughput wet-lab experiments to generate proprietary data and, in FY25, were rebranded under the AI4S Intelligent Solutions segment (Robotic Labs + Intelligent Services). The platform operates through a self-reinforcing flywheel of AI design → robotic execution → data feedback → model retraining, with AI agents now autonomously running over 10,000 compound synthesis experiments per week. The division is also increasingly commercialized and differentiated, with products such as NeoDispenser for micro-scale powder dispensing and an autonomous reaction-condition screening system deployed within a leading global pharmaceutical company's workflow. These capabilities helped drive Drug

Discovery Solutions revenue up 418.9% YoY to RMB537.9mn in FY25 from RMB103.7mn, while strengthening a proprietary data moat that software-only peers find hard to replicate.

Figure 29: XTalPi has two key facilities to house its robotic labs



Shenzhen wet lab



Shanghai wet lab

Source: Company reports.

XTalPi's incubation model converts its core AI-robotics platform into a portfolio of equity-linked ventures that act as both R&D testbeds and revenue-sharing partners. XTalPi combines investment (equity stakes) with deep R&D empowerment, staying involved from hit identification and lead optimization through IND-enabling work and, in some cases, into early clinical stages — an arrangement that lets it capture a share of pipeline upside, including commercialization proceeds in the form of royalties on incubated programs. This creates a two-way feedback loop: incubated ventures — including Signet Therapeutics (oncology), Leman Biotech (CAR-T / immunometabolic), and METiS TechBio (formerly METiS Pharmaceuticals; AI nano-delivery and RNA) — run programs on XTalPi's AI models and robotic labs and co-develop platforms with it, generating translational data that flows back to refine XTalPi's toolchains across small molecules, antibodies, and nucleic acids. The approach effectively extends XTalPi's platform into new modalities and disease areas — spanning oncology, autoimmune disease, RNA therapeutics, and nano-delivery — while the incubates bear the direct downstream clinical execution and risk, allowing XTalPi to share in the commercial upside if those pipelines succeed.

XTalPi incubating targets span different modalities, diseases and spaces

Company	Equity Stake	Business Focus	Remark
Geode	35.00%	Oncology	—
META Pharmaceuticals	15.34%	Autoimmune disease/immunometabolism	—
Signet Therapeutics	9.11%	Oncology (organoid disease models)	Invested and formed strategic collaboration in 2020
METiS TechBio	4.17%	AI-driven drug delivery (NanoForge)	Listed on HKEX Main Board, stock code 7666 HK, debuted 13 May 2026
PhoreMost Ltd.	6.67%	Oncology, targeted protein degradation	—
CytoCan Inc	14.19%	Multi-specific fusion proteins	—
ClickMab Biotech	30.00%	De novo antibody generation	—
Leman Biotech	15.26%	AI-driven tumor immunotherapy/cell therapy	Invested and established strategic collaboration in 2022
Xinshengtai (Materials Tech)	30.00%	AI-powered new materials discovery	—
Hangzhou Zentec Biotech	30.23%	Transdermal formulation R&D	—
Revir Therapeutics	Not disclosed	Rare-disease RNA therapy	RTX-117 dual China/US IND approval Jan 2026
Boundless Evolution	Not disclosed	Virtual cell platform	Strategically incubated during 2025
MorphMind	Not disclosed	No-code AI research agents	Named as incubated venture at Jan 2026 JPM HC Conference
Foundry Bio	Not disclosed	AI-designed proteins	Named as incubated venture at Jan 2026 JPM HC Conference
Jikang	Not disclosed	GPCR antibodies	Named as incubated venture at Jan 2026 JPM HC Conference
Xellar	Not disclosed	Organ-on-a-chip + AI	Named as incubated venture at Jan 2026 JPM HC Conference
Xinyue Bio	Not disclosed	DEL+AI drug discovery	Follow-on Series A+ round announced January 2026

Source: Company reports.

AI4S: XTAIPI's Platform Beyond Drug Discovery

AI4S (AI for Science) represents XtalPi's strategic evolution from a pharmaceutical-focused solutions provider into a broader scientific discovery platform. Leveraging the company's integrated capabilities in AI, quantum physics-based computation, and robotic automation, AI4S is designed to accelerate R&D across a wide range of industries. This repositioning reflects management's view that XtalPi's core technology stack functions as scalable scientific infrastructure with applications extending well beyond drug discovery.

- **The strategic importance of AI4S lies in expanding XtalPi's addressable market while reducing reliance on the lengthy and capital-intensive drug development cycle.** By deploying the same underlying technology platform across fields such as consumer health, advanced materials, and agritech, XtalPi can pursue commercial opportunities with shorter development and monetization timelines. At the same time, these applications continue to generate valuable data and model improvements that can reinforce the company's core drug discovery capabilities.

Valuations

Our PT of HK\$10 is derived from SOTP

We value XtalPi (2228.HK) via a sum-of-the-parts (SOTP) framework across five business segments, applying market EV/Sales multiples with a justified, per-segment AI premium to the four operating streams and a risk-adjusted NPV (rNPV) to its self-developed pipeline. The valuation is cross-checked with a discounted cash flow (DCF) analysis. The AI premium is anchored in documented efficiency gains and probability of success improvements, reflecting the value that the company can capture through AI-empowered R&D capabilities.

- **The SOTP framework breaking the business into 5 components.** We broke out XtalPi's sales into robotics, legacy chemical synthesis, collaboration from drug discovery, subscription from drug discovery service, and the self-developed pipeline. The first four are valued on an AI-empowered EV/Sales ratio multiplied by our 2027E sales forecast, while its self-developed pipeline is valued on probability-of-success-adjusted NPV (rNPV). The AI-empowered EV/S is derived by multiplying traditional sector EV/S with an effective AI-premium.
- **The AI premium is grounded in measurable R&D efficiency improvement.** XtalPi's prospectus documents that its platform is able to: 1) deliver crystal-structure prediction for standard small molecules in 2–3 weeks versus roughly 2 months traditionally, 2) compress hit-to-lead discovery time from ~2.5 years to <2 months, and 3) achieve 100% success rate in crystal structure prediction. We believe these translate into an AI premium multiplier of 2x-6x the traditional sector multiple.

Figure 30: With its AI platforms, XTaiPi is able to achieve substantial compression of R&D cycle and much higher hit rate in finding target compounds.

Company	Metric Category	Traditional Baseline	AI-Empowered Result
XTaiPi	Hit-to-Lead Discovery Time	~2.5 years	< 2 months
	400-Molecule Library Synthesis Time	At least 3 weeks	5 days
	Finding Active Target Compounds (Hit Rate)	8.5%	36%
	Crystal Structure Prediction (CSP) Success Rate	86% to 93%	100%
	Crystal Structure Prediction (CSP) Time for a Small Molecule	2 months	2 to 3 weeks
	Binding Pose High-Accuracy Prediction Rate	~30% (commercial software)	~56%
	AI-Driven Reaction Success Rate (Library Synthesis)	Trial-and-Error	>90%
	Chemical Reaction Prediction Accuracy	Human Expert Level	>80%
	LCMS Spectrum Analysis Accuracy	Manual Review	>95%
	X-Buddy R&D Research Cycle Time	Several weeks	30 minutes

Source: Company reports.

- **Probability of success increments are applied across both the pipeline and the contingent milestone.** On top of the traditional probability of success (PoS), we assign a theoretical AI-increment of probability of success for its self-developed pipeline assets and drug collaboration business. For example, a clinical ph.1 trial typically has 5-10% PoS. We think AI platforms will increase the PoS by 10% to 15-20%.
- **Our Dec-2027 price target of HK\$10 is based on these assumptions: 1) AI leads to 10% PoS increment in drug or product development, and 2) ~20x AI-empowered EV/S, that is at the lower end of global AIDD peers trading at 3x-148x (median ~50x).**

Figure 31: XTaiPi's SOTP Valuation

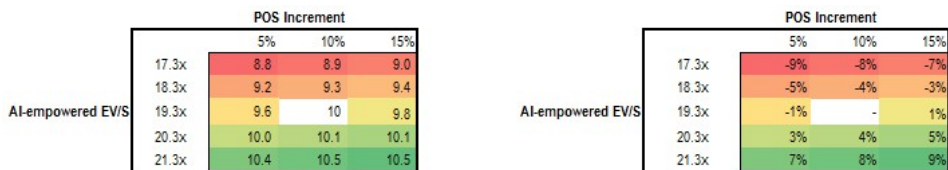
XtaiPi	Traditional		AI Premium	AI-empowered	
SoTP (Rmb mn)	27E Sales	EV/S	R&D efficiency gain	EV/S	Implied EV
Intelligence solution	669				13,619
- Robotics	254	3.2x	6.0x	19.2x	4,890
- Chem syn service	414	4.9x	4.3x	21.1x	8,728
Drug discovery	778				14,264
- Collabs	595	8.5x	2.5x	21.3x	12,648
- Subscription/FFS	183	2.1x	4.3x	8.9x	1,616
Total	1,446			19.3x	27,882
Self run assets - NPV					1,315
Net cash/(debt)					6,624
Equity value (Rmb mn)					35,822
Implied P/S (27E)					24.8x
Shares Outstanding (mn)					4,303
Per Share Value (RMB)					8.3
Per Share Value (HK\$)					10

Source: J.P. Morgan estimates.

Sensitivity analysis

- **Our sensitivity analysis has two variables: AI-empowered EV/S and PoS increment.** We flex the PoS —raising probabilities at the upper end and lowering them at the lower end —so success odds move consistently with the sector that AI contributes to drug development success. The AI-empowered EV/S is derived from multiplying AI premium with traditional sector EV/S. We note that XtalPi’s valuation, based on our SOTP analysis, is less sensitive to POS change but more sensitive to AI-empowered EV/S change. More specifically, if the multiple increases/decreases by 1 unit from 18.8x to 19 .8x/17.8x, our valuation would increase/decrease by 4%.
- If we assign no AI premium, the EV in this SOTP analysis would reduce by ~50%.

Figure 32: XtalPi’s Valuation is More Sensitive to our AI-empowered EV/S



Source: J.P. Morgan estimates.

DCF as a cross-check

The DCF serves as a triangulation cross-check on the same risk forecast. We run the DCF on the identical 2027E-anchored. The free cash flows are discounted at a 11.7% WACC. Our WACC assumes: 1) risk-free rate of 3.8%, market risk premium of 6.6%, beta of 1.2. The DCF valuation is HK\$9/shr.

Figure 33: DCF valuation

Fiscal year ends Dec 31 (All Rmb mn)	2025	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	Term. Value
P&L												
Sales	803	1,065	1,446	2,010	2,474	3,015	3,759	5,236	6,883	8,838	11,244	
COGS	(243)	(511)	(665)	(864)	(1,039)	(1,206)	(1,466)	(1,963)	(2,478)	(3,049)	(3,711)	
Gross Profit	559	554	781	1,146	1,435	1,809	2,293	3,272	4,405	5,789	7,534	
Admin exp	(410)	(443)	(465)	(488)	(512)	(538)	(565)	(593)	(623)	(654)	(687)	
R&D exp	(569)	(797)	(956)	(1,147)	(1,363)	(1,567)	(1,755)	(1,931)	(2,027)	(2,129)	(2,235)	
S&M exp	(81)	(85)	(94)	(108)	(135)	(175)	(228)	(296)	(370)	(463)	(578)	
Other income/(expenses)	556	514	559	617	669	728	800	917	1,046	1,195	1,373	
EBIT	55	(257)	(174)	19	94	256	545	1,370	2,431	3,739	5,406	
Net Interest Income (Expense)	102	103	96	95	100	109	125	151	201	284	414	
Share of Losses of Equity-accounted Investees	(15)	(15)	(15)	(15)	(15)	(15)	(15)	(15)	(15)	(15)	(15)	
Pre-tax income	135	(169)	(93)	99	179	351	656	1,506	2,617	4,008	5,806	
Tax Expense	(1)	-	-	(5)	(14)	(35)	(79)	(226)	(393)	(601)	(871)	
Minority Interest	6	11	12	13	13	14	15	15	16	17	18	
Net Income	140	(158)	(81)	107	178	329	591	1,295	2,240	3,424	4,953	
Margins												
Admin exp margin	51%	42%	32%	24%	21%	18%	15%	11%	9%	7%	6%	
R&D exp margin	71%	75%	66%	57%	55%	52%	47%	37%	29%	24%	20%	
S&M exp margin	10%	8%	6%	5%	5%	6%	6%	6%	5%	5%	5%	
EBIT margin	7%	-24%	-12%	1%	4%	9%	15%	26%	35%	42%	48%	
Tax rate	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	
Cash Flow and PV												
EBIT	55	(257)	(174)	19	94	256	545	1,370	2,431	3,739	5,406	
- Tax	(8)	-	-	(3)	(14)	(38)	(82)	(205)	(365)	(561)	(811)	
- Capex	(68)	(85)	(116)	(141)	(124)	(151)	(150)	(157)	(206)	(265)	(337)	
+ D&A	116	89	93	106	111	117	126	134	141	156	177	
Change in NWC	-	(136)	(71)	(96)	(89)	(59)	(102)	(213)	(281)	(267)	(268)	
FCF to Firm	95	(390)	(268)	(114)	(21)	125	338	928	1,721	2,801	4,168	50,719
Exponent	-	-	-	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	
Discount Factor	1.00	1.00	1.00	0.90	0.80	0.72	0.65	0.58	0.52	0.47	0.42	
PV of FCF	95	(390)	(268)	(103)	(17)	90	219	540	897	1,310	1,749	21,287
WACC Assumptions												
Risk-free rate	3.8%											
Market risk	6.6%											
Beta	1.2											
Cost of equity	11.6%											
Cost of debt	4.0%											
Liabilities as a % of EV	2.0%											
WACC	11.5%											
WACC	11.5%											
Terminal growth rate	3.0%											
Terminal Present Value	21,287											
PV Cash Flow	4,417											
Enterprise Value (RMB mn)	25,704											
YE27 Net Cash	6,624											
Equity value (RMB mn)	32,328											
Fair Value as of Dec-2027												
Per Share Value (RMB)	¥7.5											
Per Share Value (HK\$)	HKD 9											
Shares Outstanding (mn)	4,303											

Source: J.P. Morgan estimates.

Investment Thesis, Valuation and Risks

XtalPi - H (*Overweight; Price Target: HK\$10.00*)

Investment Thesis

We rate XtalPi at Overweight as a differentiated, service-led way to gain exposure to China's AI drug discovery theme. Unlike pipeline-centric peers that rely heavily on binary clinical outcomes, XtalPi generates revenue today through AI-enabled discovery services and AI-for-Science contracts, leveraging its unique robotics-driven "Physical AI" platform that integrates AI models with automated wet-lab experimentation in a closed-loop discovery engine. This scalable infrastructure creates a durable competitive moat, reinforced by proprietary experimental data and applicability beyond pharmaceuticals into materials science and consumer health. Supported by a diversified roster of blue-chip partners, including Eli Lilly, Janssen, UCB and Pfizer, and a sizable backlog, XtalPi offers a broader and lower-risk commercialization pathway, although revenue remains inherently milestone-driven and uneven. Looking ahead, we see multiple catalysts from partner milestones, asset progression and new business development opportunities, providing attractive upside while building on an already established and growing revenue base rather than depending on binary proof-of-concept events.

Valuation

Our Dec-2027 price target is HK\$10, which is based on SOTP. We break down XtalPi's business into 1) Intelligence solution, 2) drug discovery, and 3) self-developed pipeline, with the first two being valued using an assigned AI-empowered EV/S and the third business at (risk-adjusted net present value) rNPV. The implied 27E P/S is ~25x, which is at the lower end of global AIDD peers trading at 5x-173x (median ~60x).

Risks to Rating and Price Target

Key downside risks include unfavorable sentiment towards AI themes, sluggish outlicensing deals with lower-than-expected value, unfavorable clinical trial outcomes, geopolitical risks including rising US-China tension, unfavorable events against AI drug discovery sentiment, and worse-than-expected sales & earnings

XtalPi - H: Summary of Financials

Income Statement						Cash Flow Statement					
	FY24A	FY25A	FY26E	FY27E	FY28E		FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	266	803	1,065	1,446	2,010	Cash flow from operating activities	(479)	(165)	(77)	79	264
COGS	(143)	(243)	(511)	(665)	(864)	o/w Depreciation & amortization	139	116	89	93	106
Gross profit	123	559	554	781	1,146	o/w Changes in working capital	0	0	(136)	(71)	(96)
SG&A	(489)	(491)	(528)	(559)	(596)	Cash flow from investing activities	74	(3,946)	(85)	(116)	(141)
Adj. EBITDA	(545)	171	(168)	(82)	125	o/w Capital expenditure	(58)	(68)	(85)	(116)	(141)
D&A	(139)	(116)	(89)	(93)	(106)	as % of sales	21.8%	8.4%	8.0%	8.0%	7.0%
Adj. EBIT	(684)	55	(257)	(174)	19	Cash flow from financing activities	840	5,567	0	0	0
Net Interest	49	102	103	96	95	o/w Dividends paid	-	-	-	-	-
Adj. PBT	(1,515)	135	(169)	(93)	99	o/w Shares issued/(repurchased)	905	5,291	0	0	0
Tax	0	(1)	0	0	(5)	o/w Net debt issued/(repaid)	(65)	275	0	0	0
Minority Interest	(2)	(11)	(11)	(12)	(13)	Net change in cash	455	1,407	(162)	(37)	123
Adj. Net Income	(459)	247	(40)	45	242	Adj. Free cash flow to firm	(586)	(335)	(265)	(133)	33
Reported EPS	(0.35)	0.03	(0.04)	(0.02)	0.02	y/y Growth	529.0%	(42.8%)	(20.9%)	(49.6%)	(124.9%)
Adj. EPS	(0.11)	0.06	(0.01)	0.01	0.06						
DPS	-	-	-	-	-						
Payout ratio	-	-	-	-	-						
Shares outstanding	4,303	4,303	4,303	4,303	4,303						
Balance Sheet						Ratio Analysis					
	FY24A	FY25A	FY26E	FY27E	FY28E		FY24A	FY25A	FY26E	FY27E	FY28E
Cash and cash equivalents	1,315	4,484	4,322	4,285	4,409	Gross margin	46.3%	69.7%	52.0%	54.0%	57.0%
Accounts receivable	99	151	204	269	363	EBITDA margin	(204.5%)	21.3%	(15.8%)	(5.7%)	6.2%
Inventories	26	127	252	310	379	EBIT margin	(256.8%)	6.9%	(24.1%)	(12.1%)	1.0%
Other current assets	214	416	594	717	880	Net profit margin	(172.1%)	30.8%	(3.8%)	3.1%	12.0%
Current assets	3,315	7,467	7,483	7,569	7,855	ROE	(23.1%)	3.7%	(0.4%)	0.5%	2.5%
PP&E	320	269	285	319	364	ROA	(21.1%)	3.4%	(0.4%)	0.4%	2.3%
LT investments	555	2,096	2,096	2,096	2,096	ROCE	(34.1%)	0.8%	(2.6%)	(1.8%)	0.2%
Other non current assets	711	2,248	2,239	2,238	2,238	SG&A/Sales	183.5%	61.2%	49.6%	38.6%	29.6%
Total assets	4,355	10,217	10,230	10,339	10,661	Net debt/Equity	NM	NM	NM	NM	NM
Short term borrowings	52	171	171	171	171	Net debt/EBITDA	2.3	NM	23.6	48.2	NM
Payables	16	62	84	109	137	Sales/Assets (x)	0.1	0.1	0.1	0.1	0.2
Other short term liabilities	212	272	292	319	359	Assets/Equity (x)	1.1	1.1	1.1	1.1	1.1
Current liabilities	280	506	547	599	667	Interest cover (x)	11.1	NM	1.6	0.8	NM
Long-term debt	0	175	175	175	175	Operating leverage	(10.0%)	(53.7%)(1728.3%)	(89.7%)	(284.8%)	
Other long term liabilities	83	281	281	281	281	Tax rate	0.0%	0.6%	0.0%	0.0%	5.0%
Total liabilities	362	787	828	880	948	Revenue y/y Growth	52.8%	201.2%	32.7%	35.8%	39.0%
Shareholders' equity	3,964	9,384	9,343	9,388	9,630	EBITDA y/y Growth	(24.6%)	(131.4%)	(198.6%)	(51.5%)	(252.6%)
Minority interests	29	47	59	70	83	EPS y/y Growth	-	(153.9%)	(116.4%)	(211.4%)	436.2%
Total liabilities & equity	4,355	10,217	10,230	10,339	10,661						
BVPS	0.92	2.18	2.17	2.18	2.24						
y/y Growth	-	136.7%	(0.4%)	0.5%	2.6%						
Net debt/(cash)	(1,263)	(4,138)	(3,976)	(3,940)	(4,063)						

Source: Company reports and J.P. Morgan estimates.

Note: Rmb in millions (except per-share data). Fiscal year ends Dec. o/w - out of which

Initiation

Overweight

3696.HK, 3696 HK

Price (23 Jul 26): HK\$48.44

Price Target (Dec-27): HK\$71.00

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Insilico - H

Deep AIDD Pipeline with Novel Targets and Continued Out-Licensing Potential; Initiate at OW

Insilico is the novel-target, pipeline-deep way to own the China AIDD theme, and we initiate at Overweight. Different from XtalPi, its business mostly focuses on drug out-licensing, underpinned by upfront-plus-milestone economics and marquee partners such as Eli Lilly, which represents the economics of early drug molecules via AI-enabled R&D. What sets Insilico apart is its vertical depth rather than horizontal breadth: it has advanced Rentosertib (TNIK [TRAF2 and NCK-interacting kinase] targeted) for IPF (idiopathic pulmonary fibrosis) into Phase III—among the most clinically advanced drugs where AI identified both a novel target and designed the molecule (de novo AIDD), a distinction from AI-optimized drugs advanced by many AIDD peers. As a drug originator, it has built a 30-asset pipeline and 10+ partnerships worth US\$7.5bn in total contract value, headlined by the US\$2.75bn Lilly collaboration. While XtalPi offers services-led breadth, Insilico offers deeper, milestone-driven torque from a large pipeline—an equally compelling but distinct growth story within the pair.

- **Insilico's novel-target clinical lead is a hard-to-replicate moat.** Rentosertib is among the most advanced de novo AIDD assets globally—an AI-discovered TNIK inhibitor that showed encouraging FVC preservation in Phase IIa and is now entering Phase III GENESIS-IPF-3, with first-patient-in during 2026 and topline data by 2029. Its differentiation rests on AI identifying a previously undrugged target, distinct from peers such as Generate:Biomedicines or Schrödinger whose Phase III-stage assets are AI-optimized against established targets. We note, however, that the moat ultimately rests on forthcoming Phase III data to confirm efficacy, and the platform spans the entire Pharma.AI stack—PandaOmics target discovery, Chemistry42 generative design, inClinico trial prediction—at roughly 12–18 months and US\$3–5mn per preclinical candidate, a data flywheel that compounds the lead over time.
- **A high-value out-licensing flywheel drives repeatable monetization.** The platform has produced a 30-asset pipeline and 10+ deals worth US\$7.5bn in aggregate value, with ~US\$205mn realized by 2025 and US\$7.3bn in potential future milestones and royalties. The model is repeatable—software, to collaboration, to investment, to out-licensing—best shown by the US\$2.75bn Lilly deal (US\$115mn upfront, up to US\$2.64bn milestones) plus out-licensing of USP1, KAT6, KIF18A and GLP-1R. Because each deal scales revenue without proportional capex, Insilico funds a deep pipeline while preserving runway. In our model, we see some assets like Rentosertib and garutadustat to generate US\$2.5–3.0bn peak sales and Insilico to benefit from royalties and milestones.
- **A dense catalyst calendar plus new platform optionality underpins re-rating.** Near-term value-inflection events span 2026–2027: the PHD12 (garutadustat) Phase IIa BETHESDA IBD readout by 2027, early efficacy data from pan-TEAD ISM6331, GLP-1R out-license economics, and 8–10 new preclinical candidates targeted in 2026. Layered on top is the emerging Science MMAI Gym (with a Liquid AI partnership), opening a new recurring-revenue vector beyond the pipeline. Each catalyst is a discrete, high-magnitude trigger, reinforcing Insilico as the milestone-driven complement to XtalPi.
- **Valuation & Risk:** Our Dec-2027 PT of HK\$71 is based on a Sum-Of-the-Parts methodology. Key downside risks include unfavorable sentiment towards AI themes, slug-

gish outlicensing deals with lower-than-expected value, unfavorable clinical trial outcomes, geopolitical risks including rising US-China tension, and worse-than-expected sales & earnings.

Price Performance



Company Data

Shares O/S (mn)	579
52-week range (HK\$)	80.90-29.98
Market cap (\$ mn)	3,576
Exchange rate	7.84
Free float (%)	28.6%
3M ADV (mn)	9.53
3M ADV (\$ mn)	52.7
Volatility (90 Day)	95
Index	HSI
BBG ANR (Buy Hold Sell)	9 0 0

Key Metrics (FYE Dec)

\$ in millions	FY25A	FY26E	FY27E	FY28E
Financial Estimates				
Revenue	56	192	280	1,504
Adj. EBITDA	(60)	23	66	1,130
Adj. EBIT	(65)	21	62	1,123
Adj. net income	(44)	42	85	1,150
Adj. EPS	(0.08)	0.08	0.15	2.06
BBG EPS	(0.07)	(0.02)	0.23	0.43
Cashflow from operations	(77)	4	68	913
FCFF	(80)	(7)	51	838
Margins and Growth				
Revenue Growth Y/Y (%)	(34.5%)	242.1%	45.5%	437.2%
EBITDA margin	(106.3%)	11.9%	23.5%	75.1%
EBITDA Growth Y/Y (%)	90.0%	(138.3%)	186.8%	1619.2%
EBIT margin	(114.8%)	11.1%	22.2%	74.7%
Net margin	(77.9%)	22.0%	30.3%	76.4%
Adj. EPS growth	-	(196.5%)	100.7%	1255.1%
Ratios				
Adj. tax rate	(0.0%)	0.0%	0.0%	0.0%
Interest cover	NM	114.5	328.5	5,647.5
Net debt/Equity	NM	NM	NM	NM
Net debt/EBITDA	6.6	NM	NM	NM
ROCE	(23.3%)	4.5%	11.6%	97.3%
ROE	41.4%	8.9%	15.8%	99.6%
Valuation				
FCFF yield	(2.3%)	(0.2%)	1.5%	24.3%
Dividend yield	-	-	-	-
EV/Revenue	52.7	15.4	10.4	1.4
EV/EBITDA	NM	129.6	44.4	1.8
Adj. P/E	NM	81.7	40.7	3.0

Summary Investment Thesis and Valuation

Investment Thesis

We rate Insilico Medicine at Overweight as the novel-target, pipeline-led way to gain exposure to China's AI drug discovery theme. While sharing the same AI-enabled drug R&D tailwinds and milestone-driven business model as XtalPi, Insilico differentiates itself through deeper vertical integration and ownership of a high-value proprietary pipeline, led by Rentosertib, one of the world's most advanced de novo AI-discovered drug candidates to enter Phase III development. Supported by its end-to-end Pharma.AI platform spanning target discovery, molecule design and clinical trial prediction, Insilico has built a 30-asset pipeline and secured more than US \$7.5bn of partnership value, including its landmark collaboration with Eli Lilly. Its repeatable out-licensing model provides a scalable monetization engine, generating upfront payments, milestones and future royalties while funding continued pipeline expansion. With multiple clinical readouts, business development opportunities and new platform initiatives expected over 2026–2027, we see a rich catalyst path that offers significant upside, albeit with higher clinical and execution risk than service-led peers given its greater dependence on proprietary pipeline success.

Valuation

Our Dec-2027 price target is HK\$71, which is based on SOTP. We broke down Insilico's business into 1) drug discovery, 2) software solution & others. They are being valued using an assigned AI-empowered EV/S. The implied 27E P/S is ~19x, which is at the lower end of global AIDD peers trading at 5x-173x (median ~60x).

Potential of Insilico's Deep Pipeline

Figure 34: Insilico's Sales Build

Sales build	2022A	2023A	2024A	2025A	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E
USD in Millions														
TOTAL REVENUE	30.1	51.2	85.8	56.2	192.4	208.8	1,584.3	433.1	1,180.8	1,562.3	826.5	713.6	1,197.8	1,628.9
Drug Discovery	25.8	46.8	79.7	48.8	184.1	270.8	1,493.9	421.4	1,167.6	1,547.5	889.7	694.5	1,176.2	1,604.4
Out-licensed royalty revenue (risk-adj.)	-	-	-	-	-	-	-	2.3	25.7	141.1	229.8	325.4	424.3	518.3
Upfront & milestone revenue (risk-adj.)	25.3	45.0	75.6	23.9	154.0	235.8	1,450.6	366.1	1,078.2	1,329.5	486.8	256.7	616.3	321.5
Drug discovery services revenue	0.5	1.0	3.1	25.0	30.1	35.8	43.3	53.1	63.7	76.9	93.0	112.4	135.5	164.6
Software solution revenue	2.8	3.4	4.0	4.9	5.7	6.5	7.5	8.7	10.0	11.5	13.3	15.4	17.7	20.5
Other discovery revenue	1.5	1.8	2.1	2.5	2.6	2.7	2.9	3.0	3.2	3.3	3.5	3.7	3.9	4.1
YoY														
TOTAL REVENUE	78%	68%	-34%	242%	48%	437%	-71%	173%	32%	-47%	-14%	68%	-14%	
Drug Discovery	78%	73%	-39%	277%	47%	452%	-72%	177%	33%	-48%	-14%	69%	-15%	
Out-licensed royalty revenue (risk-adj.)										449%	63%	42%	30%	22%
Upfront & milestone revenue (risk-adj.)		77%	70%	-69%	545%	53%	517%	-75%	195%	23%	-63%	-47%	140%	48%
Drug discovery services revenue		100%	214%	694%	21%	19%	21%	23%	20%	21%	21%	21%	20%	22%
Software solution revenue		21%	17%	24%	16%	15%	15%	15%	15%	15%	15%	15%	15%	15%
Other discovery revenue		20%	18%	17%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%

Source: J.P. Morgan estimates, Company data.

Rentosertib (TNIK inhibitor): Most-Advanced Pipeline Asset with AI-enabled Novelty in Targeting IPF

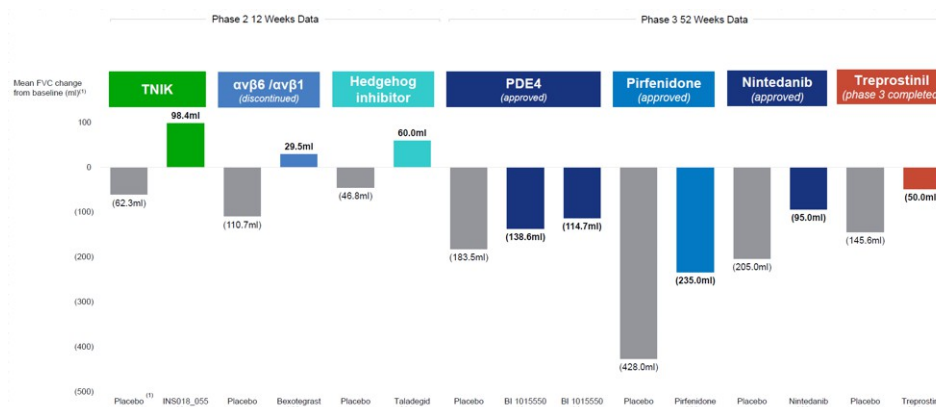
- Rentosertib (ISM001-055), Insilico Medicine's lead asset for Idiopathic Pulmonary Fibrosis (IPF), stands out in the biotech landscape not merely for its rapid development, but for the fundamental way its biological target was selected.** By leveraging generative AI to uncover a completely novel therapeutic target, Rentosertib serves as the definitive proof-of-concept that artificial intelligence can successfully guide target selection, unlocking previously unexplored biological pathways. In our model, we forecast the peak global sales for the drug to be US\$3bn.
- AI-driven target discovery identified a causal disease mechanism that traditional human research has historically overlooked.** Utilizing its PandaOmics platform, Insilico analyzed massive multi-omics datasets, scientific literature, and clinical trial text data to pinpoint TNIK as a highly confident therapeutic target for IPF. While TNIK was known to be associated with key fibrosis-driving pathways (such as WNT, TGF- β , and NF- κ B), it had never been studied as a therapeutic target for IPF until the AI algorithm explicitly selected it based on causal inference and biological network scoring.
- The seamless integration of AI-driven target discovery and molecular design validates the industry's first end-to-end generative pipeline.** After the AI identified the novel TNIK target, Insilico utilized its Chemistry42 generative platform to design the specific small molecule to inhibit it, creating a highly selective compound with desirable drug-like properties. This dual-engine approach makes Rentosertib the first drug in industry history where both the novel biological target and the chemical structure were entirely discovered and designed by AI, leading to its historic United States Adopted Name (USAN) designation.
- Phase IIa clinical efficacy is de-risking and appears much better than the Ph3 efficacy of recently FDA approved nerandomilast.** The true differentiator of rentosertib is that the AI's novel target hypothesis successfully translated into human clinical efficacy in a fraction of traditional development timelines, reaching Phase I trials in just 30 months. In a double-blind Phase IIa trial published in Nature Medicine, IPF patients receiving rentosertib 60mg achieved a mean improvement in lung function (+98.4 mL forced vital capacity [FVC])

compared to a decline in the placebo group (-20.3 mL), proving that the AI's biological target selection was clinically relevant and commercially viable. However, we note patients concurrently taking 60 mg rentosertib QD with either nintedanib or pirfenidone did not exhibit significant changes in FVC, while patients receiving 60 mg rentosertib QD not concurrently taking SOC antifibrotic therapy exhibited significant improvement in FVC (+187.8 ml).

Patients who received 30 mg rentosertib QD showed a mean reduction in FVC of -27.0 ml, patients who received 30 mg rentosertib BID experienced a mean change of +19.7 ml. As a comparison, in the Ph3 trial of nerandomilast, the adjusted mean changes in FVC at week 52 were -114.7 ml in the nerandomilast 18-mg group, -138.6 ml in the nerandomilast 9-mg group, and -183.5 ml in the placebo group. Although the Ph2 efficacy of AK3280 appears better than that of rentosertib, we think the existing mechanism of AK3280, an optimized version of pirfenidone, could cap the upside potential of AK3280's Ph3 data. In Ph2, the 400 mg group of AK3280 achieved particularly impressive results, with an FVC increase of 209.4 ml and a 6.4% adjusted percent predicted FVC improvement from baseline.

- **Rentosertib is a selective oral TNIK inhibitor for idiopathic pulmonary fibrosis (IPF) and the world's most clinically advanced AI-discovered, AI-designed drug candidate.** The target was identified using PandaOmics, while the molecule was designed using Chemistry42. Following a positive Phase IIa readout, the program has advanced into the pivotal Phase III GENESIS-IPF-3 trial in China, initiated in July 2026, with topline data expected around 2029 per company data.
- **The Phase III study narrows development to the winning 60 mg once-daily (QD) dose and extends treatment duration to 52 weeks,** transitioning from proof-of-concept efficacy to registrational confirmation. The earlier Phase IIa (GENESIS-IPF) was a 12-week, 71-patient, 22-site dose-ranging study in China, where the 60 mg QD arm demonstrated +98.4 mL FVC improvement versus -20.3 mL on placebo. We think the Phase III design signals confidence, as patients remain on standard-of-care antifibrotics. As a result, Rentosertib must demonstrate efficacy on top of existing treatment with nintedanib or pirfenidone.
- **IPF remains a high-need and underserved market, with only three drugs approved globally as of July 2026:** Pirfenidone (Esbriet, Roche), Nintedanib (OFEV, Boehringer Ingelheim), Nerandomilast (Jascayd, Boehringer Ingelheim) PDE4B inhibitor; FDA approval in October 2025). Nintedanib generics are expected to enter China in 2026. Existing therapies primarily slow disease progression and are associated with tolerability limitations, while several high-profile candidates, including pamrevlumab, bexotegrast, and GLPG1690, have failed in development. Rentosertib's novel oral, first-in-class TNIK mechanism is differentiated from the crowded TGF- β /kinase landscape. By simultaneously modulating the Wnt, TGF- β , and NF- κ B pathways, the drug benefits from both Breakthrough Therapy Designation and FDA Orphan Drug Designation, creating a meaningful competitive advantage.

Figure 35: Rentosertib demonstrated clear efficacy superiority against other competing drugs.



Source: Company reports.

- Rentosertib is Insilico's clearest platform-validation asset and arguably its most attractive out-licensing candidate.** However, management has chosen to retain ownership of the core IPF opportunity while monetizing adjacent indications. Insilico reportedly carved IPF and fibrosis assets out of its Sanofi collaboration to preserve the franchise's strategic value. Additional label-expansion opportunities include: Kidney fibrosis (IND submission stage), Inhaled Rentosertib (IND submission stage). A successful Phase III readout around 2029 would represent the first true clinical validation of an AI-discovered drug and could unlock a landmark ex-China partnership for Rentosertib itself.

ISM5411 (Garutadustat, PHD1/2 inhibitor): First-in-Class Gut-Restricted PHD Inhibitor for IBD

- ISM5411 (garutadustat) is Insilico's wholly-owned, potentially first-in-class oral gut-restricted PHD1/2 inhibitor for inflammatory bowel disease,** which one of the platform's clearest demonstrations of AI-driven speed and a key near-term catalyst as its Phase IIa trial progresses. In our model, we forecast its peak sales to reach ~US\$2bn.
- ISM5411 is an AI-discovered IBD asset, of which Insilico has full economic capture and concentrated upside.** Discovered end-to-end on Pharma.AI with its preclinical candidate, was nominated in November 2021, it targets ulcerative colitis (UC) and Crohn's disease. Because no partner economics are attached, the company retains the entire value of the program if it succeeds—making it a high-torque item within the pipeline.
- ISM5411's differentiation lies in gut-restricted, PHD1/2-selective inhibition that drives local barrier repair while limiting systemic exposure.** It inhibits PHD1/2 with IC50 of ~1.9 nM against PHD1 and ~2.5 nM against PHD2 (with minimal PHD3 activity), stabilizing HIF-1α to induce gut-barrier protective genes and anti-inflammation. A high colon-to-plasma ratio concentrates drug exposure at the site of disease, underpinning its potential first-in-class positioning and suitability for combination with existing IBD therapies.
- ISM5411 has advanced from positive Phase I into an actively enrolling**

Phase IIa, its critical de-risking readout, with early franchise validation already in hand. Insilico completed two Phase I studies in Australia and China—including a first-in-human Australia study in n=76 healthy volunteers—with positive topline reported in January 2025 showing the drug was safe, well-tolerated and displayed gut-restrictive PK. It has since advanced into the BETHESDA Phase IIa (NCT072655), a multicenter, randomized, double-blind, placebo-controlled study in approximately 80 UC patients led by Prof. Minhu Chen of the First Affiliated Hospital of Sun Yat-sen University, with first-patient-first-dose completed on January 12, 2026; endpoints span safety, tolerability, PK, and efficacy via clinical remission/response, endoscopy, histopathology, and biomarkers.

- **Garutadustat's competitive thesis rests entirely on the mechanism and route differentiation, not yet on data.** Its non-immunosuppressive, gut-restricted barrier-repair mechanism is orthogonal to the immunosuppressive biologics and JAKs that dominate the IBD market, and its oral, combination-friendly design targets a niche as an add-on rather than a replacement.
- **The BETHESDA Phase IIa readout (~2027) is the pivotal value-inflection event that could increase the chance of a large global licensing opportunity.** Because garutadustat has no disclosed efficacy data yet, its ex-China BD value is currently option-like; a positive 2027 efficacy signal on remission/endoscopic endpoints would materially de-risk it and support a higher-value out-licensing or co-development transaction.

Figure 36: Insilico's AI modules act together to form a robust drug discovery platform

Pharma.AI Module	Analogy	Core AI Function	Key Output & Benefit
Biology42	The "Disease Detective"	Target discovery and prioritization.	Identifies novel biological targets and mechanisms from massive omics data to provide a precise rationale for drug development.
Chemistry42	The "Master Chemist"	Small molecule generation and optimization.	Rapidly invents, screens, and optimizes new chemical entities to identify candidates with high developmental potential.
Medicine42	The "Virtual Crystal Ball"	Clinical trial outcome prediction.	Forecasts Phase II to Phase III conversion success, proactively identifying risks and de-risking expensive human trials.
Science42	The "Research Assistant"	Scientific document drafting.	Automates the synthesis and drafting of academic papers and regulatory documents, significantly accelerating research efficiency.

Source: Company reports.

Figure 37: Insilico has significantly expanded outlicensing in 2026

Date	Partner	Asset/Target	Type	Upfront	Total Value
Jun-21	Teva	Target discovery (PandaOmics)	Research collaboration	Undisclosed	Undisclosed
Aug-21	Avinas	AI-driven PROTAC R&D	Research collaboration	Undisclosed	Undisclosed
Jan-22	Fosun Pharma	Four targets	Strategic AI drug-discovery collaboration	Undisclosed	>US\$100m + royalties
Nov-22	Sanofi	Up to 6 targets (Pharma.AI)	Target discovery + out-licensing	Up to US\$21.5m (upfront + target-nomination fees)	Up to US\$1.2bn + royalties
Sep-23	Exelixis	ISM3091 (USP1)	Out-licensing	US\$80m	~US\$0.98bn + royalties
2023	Menarini / Stemline	ISM5043 (KAT6)	Out-licensing	US\$12m	>US\$500m + royalties
2024	Menarini / Stemline	ISM9682 (KIF18A)	Out-licensing	US\$20m (+US\$8m milestone)	>US\$550m + royalties
2026	Hygia Therapeutics	ISM8969 (NLRP3)	Co-development (50/50)	Up to US\$66m (upfront + milestones)	Up to US\$66m
Jan-26	Servier	Oncology programs	Research collaboration	US\$32m (upfront + near-term)	>US\$888m + royalties
Jan-26	Qilu Pharmaceutical	Cardiometabolic targets	Strategic collaboration	Undisclosed	~US\$120m
Mar-26	Eli Lilly	Oral small-molecule portfolio	Global AI-driven R&D collaboration + out-licensing	US\$115m	Up to US\$2.75bn + royalties
Mar-26	Tenacia Biotechnology	CNS targets (expanded)	Research collaboration	Undisclosed	Up to US\$94.75m
Jun-26	SK Biopharmaceuticals	CNS / neuroimmune	R&D collaboration + out-licensing (largest APAC deal)	US\$18m (upfront/near-term)	>US\$2.5bn
Jul-26	Bora Pharmaceuticals	Multi-target (AI discovery + CDMO)	Strategic alliance (manufacturing, non-binding)	Undisclosed	>US\$2.5bn

Source: Company reports.

Valuations

Our PT of HK\$71 is derived from SOTP

We value Insilico Medicine (3696.HK) on a sum-of-the-parts (SOTP) basis. As a pipeline-deep AIDD company, its valuation is largely anchored by its collaboration revenue, with AI service streams as a secondary contributor. Our analysis yields HK\$71 per share. The key to Insilico's premium lies in demonstrated clinical validation—superior success rates and predictive accuracy, allowing us to justify a higher pipeline probability of success (PoS) versus traditional biotech.

- Pipeline-led value shifts focus to clinical probabilities over platform multiples.**
The collaboration revenue accounts for about ~US\$4.4bn of its total EV of ~US\$5bn, making the PoS assumption the primary driver of valuation. This dynamic is structurally different from breadth-first platforms like XTAIPI, where it has meaningful amount of service businesses. Here, valuation is more sensitive to clinical readouts, particularly a few lead assets.
- Premium is supported by differentiated clinical-outcome evidence.**
Insilico reports Phase I success rates of 80–90% versus a traditional 40–65% baseline, and Phase I-to-approval probability of 9–18% versus 5–10% historically. This suggests that the platform enhances probability of success, not just development speed—an important distinction underpinning our premium thesis and one that is difficult for peers to replicate.
- Predictive-accuracy metrics provide auditable support for the premium.**
The company discloses measurable benchmarks, including 0.86 AUC-ROC for Phase II outcome prediction (inClinico), ~84% protein-folding accuracy, and ~75% toxicity-prediction accuracy. These quantitative indicators serve as evidence that Insilico's AI confers a real, reproducible edge beyond narrative positioning. We apply a uniform PoS uplift of 15% (vs XAIPI's 10%) to reflect improved clinical probabilities.

Figure 38: Remarkably, Insilico is able to reduce time to pre-clinical candidate from 4.5 years to 12-18 months, thanks to its AI platform.

Company	Metric Category	Traditional Baseline	AI-Empowered Result
Insilico Medicine	Time to Pre-Clinical Candidate (PCC)	4.5 years	12 to 18 months
	Phase I Clinical Success Rate	40% to 65%	80% to 90%
	Overall Probability of Success (Phase I to Approval)	5% to 10%	9% to 18%
	Average Cost per PCC	Traditional high costs	US\$3 million to US\$5 million
	Phase II Trial Outcome Prediction (inClinico)	N/A	0.86 AUC-ROC
	Protein Folding Prediction Accuracy	N/A	84%
	Drug Toxicity Prediction Accuracy	N/A	75%
	TargetPro Novel Target Druggability	Public platform baselines	86.5%
	TargetPro Target Retrieval Rate	Open Targets benchmark	71.6% (2-3x improvement)
	Science MMAI Gym Performance Boost	Baseline Foundation Model	Up to 10X increase

Source: Company reports.

- AI platform premium is applied selectively across revenue streams.**
We assign differentiated EV/Sales uplifts by segment: collaboration sees the largest expansion (8.8x to 15.7x), while subscription and software remain more directly driven by efficiency gains from AI (2.0x to 8.9x).
- Our Dec-2027 price target is HK\$71, based on the assumptions: 1) AI leads to 15% PoS increment in drug or product development, and 2) 17.2x AI-**

empowered EV/S, that is at the lower end of global AIDD peers trading at 3x-148x (median ~50x).

- If we assign no AI premium, the EV in this SOTP analysis would reduce by ~50%.

Figure 39: Insilico's SOTP valuation

Insilico	Traditional		AI Premium	AI-empowered	
SoTP (US mn)	27E Sales	EV/S	R&D efficiency gain	EV/S	Implied EV
Drug discovery	271			17.5x	4,737
Collaboration	235	8.8x	2.1x	18.8x	4,417
Subscription	36	2.0x	4.5x	8.9x	320
Software solution & others	9	2.0x	4.5x	8.9x	83
Total	280			17.2x	4,820
Net cash/(debt)					433
Equity value (US mn)					5,253
Implied P/S (27E)					18.8x
Shares Outstanding (mn)					572
Per Share Value (USD)					9.2
Per Share Value (HK\$)					71

Source: J.P. Morgan estimates.

Sensitivity Analysis

Our sensitivity analysis has two variables: AI-empowered EV/S and PoS increment.

We flex the PoS —raising probabilities at the upper end and lowering them at the lower end—so odds of success move consistently with the sector story that AI contributes to drug development success. The AI-empowered EV/S is derived by multiplying the effective AI premium with traditional sector EV/S. We note that, if PoS increment increases/decreases by 5%, our valuation will increase/decrease by 4%, while AI-empowered EV/S increases/decreases by 1%, our valuation will increase/decrease by 5%.

Figure 40: Insilico's sensitivity analysis

		POS Increment		
		10%	15%	20%
AI-empowered EV/S	15.2x	60.9	63.5	66.1
	16.2x	64.6	67.3	70.1
	17.2x	68.2	71	74.0
	18.2x	71.8	74.9	78.0
	19.2x	75.5	78.7	82.0

		POS Increment		
		10%	15%	20%
AI-empowered EV/S	15.2x	-14%	-11%	-7%
	16.2x	-9%	-5%	-1%
	17.2x	-4%	-	4%
	18.2x	1%	5%	10%
	19.2x	6%	11%	15%

Source: J.P. Morgan estimates.

DCF as a cross-check

The DCF serves as a triangulation cross-check on the same risk forecast. We run the DCF on the identical 2027E-anchored. The free cash flows are discounted at a 12.4% WACC. Our WACC assumes: 1) a risk-free rate of 3.8%, a market risk premium of 6.6%, and a beta of 1.3. The DCF valuation is HK\$71/shr.

Figure 41: DCF Valuation

(In USD million)	2025	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	Term. Valu
P&L												
Sales	56	192	280	1,504	433	1,181	1,562	827	714	1,198	1,029	
COGS	(10)	(25)	(31)	(150)	(41)	(106)	(133)	(66)	(54)	(84)	(67)	
Gross Profit	46	167	249	1,354	392	1,075	1,430	760	660	1,114	962	
G&A	(17)	(21)	(26)	(31)	(35)	(39)	(40)	(42)	(45)	(47)	(49)	
R&D	(81)	(106)	(138)	(172)	(206)	(248)	(285)	(328)	(360)	(396)	(416)	
S&M	(6)	(19)	(24)	(27)	(30)	(33)	(36)	(40)	(44)	(46)	(48)	
Other income/(expenses)	(5)	-	-	(0)	(0)	-	(0)	-	-	0	0	
EBIT	(65)	21	62	1,123	121	755	1,068	350	211	625	448	
Net Interest Income (Expense)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	
Other gains/(losses)	(287)	14	14	15	45	58	81	115	135	147	169	
Pre-tax income	(352)	35	76	1,138	165	813	1,148	466	346	771	617	
Tax Expense	(0)	-	-	-	-	(41)	(115)	(56)	(52)	(116)	(92)	
Net Income	(352)	35	76	1,138	165	773	1,034	410	294	656	524	
Margins												
G&A Margin	31%	11%	9%	2%	8%	3%	3%	5%	6%	4%	5%	
R&D Margin	145%	55%	49%	11%	48%	21%	18%	40%	50%	33%	40%	
S&M margin	11%	10%	8%	2%	7%	3%	2%	5%	6%	4%	5%	
Tax rate	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	
Cash Flow and PV												
EBIT	(65)	21	62	1,123	121	755	1,068	350	211	625	448	
- Tax	-	(2)	(5)	(20)	(8)	(21)	(25)	(17)	(13)	(24)	(21)	
- Capex	(3)	(12)	(17)	(75)	(22)	(47)	(47)	(25)	(21)	(36)	(31)	
+ D&A	5	1	3	6	20	20	26	30	29	27	29	
Change in NWC	(28)	(40)	(21)	(243)	214	(123)	(36)	120	22	(58)	8	
FCF to Firm	(90)	(31)	23	791	325	585	986	458	227	534	433	5,192
Exponent	-	-	-	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	
Discount Factor	1.00	1.00	1.00	0.89	0.79	0.70	0.63	0.56	0.50	0.44	0.39	
PV of FCF	(90)	(31)	23	704	257	412	618	256	113	236	170	2,041
WACC Assumptions												
Risk-free rate	3.8%											
Market risk	6.6%											
Beta	1.3											
Cost of equity	12.4%											
Cost of debt	5.0%											
Liabilities as a % of EV	0.0%											
WACC	12.4%											
WACC	12.4%											
Terminal growth rate	3.0%											
Terminal Present Value	2,041											
PV Cash Flow	2,789											
Enterprise Value (USD mn)	4,830											
YE27 Cash and Equivalents	437											
Debts	(4)											
Equity value (USD mn)	5,263											
Fair Value as of Dec-2027												
Per Share Value (USD)	9.2											
Per Share Value (HK\$)	71											
Shares Outstanding (fully diluted)	572											

Source: J.P. Morgan estimates.

Investment Thesis, Valuation and Risks

Insilico - H (Overweight; Price Target: HK\$71.00)

Investment Thesis

We rate Insilico Medicine at Overweight as the novel-target, pipeline-led way to gain exposure to China's AI drug discovery theme. While sharing the same AI-enabled drug R&D tailwinds and milestone-driven business model as XtalPi, Insilico differentiates itself through deeper vertical integration and ownership of a high-value proprietary pipeline, led by Rentosertib, one of the world's most advanced de novo AI-discovered drug candidates to enter Phase III development. Supported by its end-to-end Pharma.AI platform spanning target discovery, molecule design and clinical trial prediction, Insilico has built a 30-asset pipeline and secured more than US\$7.5bn of partnership value, including its landmark collaboration with Eli Lilly. Its repeatable out-licensing model provides a scalable monetization engine, generating upfront payments, milestones and future royalties while funding continued pipeline expansion. With multiple clinical readouts, business development opportunities and new platform initiatives expected over 2026–2027, we see a rich catalyst path that offers significant upside, albeit with higher clinical and execution risk than service-led peers given its greater dependence on proprietary pipeline success.

Valuation

Our Dec-2027 price target is HK\$71, which is based on SOTP. We broke down Insilico's business into 1) drug discovery, 2) software solution & others. They are being valued using an assigned AI-empowered EV/S. The implied 27E P/S is ~19x, which is at the lower end of global AIDD peers trading at 5x-173x (median ~60x).

Risks to Rating and Price Target

Key downside risks include unfavorable sentiment towards AI themes, sluggish outlicensing deals with lower-than-expected value, unfavorable clinical trial outcomes, geopolitical risks including rising US-China tension, unfavorable events against AI drug discovery sentiment, and worse-than-expected sales & earnings.

Insilico - H: Summary of Financials

Income Statement						Cash Flow Statement					
	FY24A	FY25A	FY26E	FY27E	FY28E		FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	86	56	192	280	1,504	Cash flow from operating activities	(57)	(77)	4	68	913
COGS	(8)	(10)	(25)	(31)	(150)	o/w Depreciation & amortization	6	5	1	3	6
Gross profit	78	46	167	249	1,354	o/w Changes in working capital	(33)	(28)	(40)	(21)	(243)
SG&A	(23)	(24)	(40)	(49)	(59)	Cash flow from investing activities	7	(50)	(12)	(17)	(75)
Adj. EBITDA	(31)	(60)	23	66	1,130	o/w Capital expenditure	(1)	(3)	(12)	(17)	(75)
D&A	(6)	(5)	(1)	(3)	(6)	as % of sales	1.1%	5.3%	6.0%	6.0%	5.0%
Adj. EBIT	(38)	(65)	21	62	1,123	Cash flow from financing activities	(2)	394	0	0	0
Net Interest	(0)	(0)	(0)	(0)	(0)	o/w Dividends paid	-	-	-	-	-
Adj. PBT	(17)	(352)	35	76	1,138	o/w Shares issued/(repurchased)	0	275	0	0	0
Tax	(0)	(0)	0	0	0	o/w Net debt issued/(repaid)	(2)	(2)	0	0	0
Minority Interest	-	-	-	-	-	Net change in cash	(51)	267	(8)	51	838
Adj. Net Income	(23)	(44)	42	85	1,150	Adj. Free cash flow to firm	(58)	(80)	(7)	51	838
Reported EPS	-	(0.63)	0.06	0.14	2.04	y/y Growth	(61533.2%)	38.0%	(90.8%)	(795.0%)	1537.0%
Adj. EPS	-	(0.08)	0.08	0.15	2.06						
DPS	-	-	-	-	-						
Payout ratio	-	-	-	-	-						
Shares outstanding	-	559	559	559	559						
Balance Sheet						Ratio Analysis					
	FY24A	FY25A	FY26E	FY27E	FY28E		FY24A	FY25A	FY26E	FY27E	FY28E
Cash and cash equivalents	126	393	386	437	1,275	Gross margin	90.4%	81.5%	87.0%	89.0%	90.0%
Accounts receivable	7	27	63	92	474	EBITDA margin	(36.7%)	(106.3%)	11.9%	23.5%	75.1%
Inventories	-	-	-	-	-	EBIT margin	(43.7%)	(114.8%)	11.1%	22.2%	74.7%
Other current assets	7	27	63	92	474	Net profit margin	(26.4%)	(77.9%)	22.0%	30.3%	76.4%
Current assets	133	474	503	583	1,803	ROE	6.8%	41.4%	8.9%	15.8%	99.6%
PP&E	7	7	17	31	100	ROA	(31.5%)	(13.8%)	8.3%	14.8%	90.8%
LT investments	-	-	-	-	-	ROCE	(74.1%)	(23.3%)	4.5%	11.6%	97.3%
Other non current assets	3	8	8	8	8	SG&A/Sales	26.8%	42.2%	20.9%	17.7%	3.9%
Total assets	144	490	529	622	1,911	Net debt/Equity	NM	NM	NM	NM	NM
Short term borrowings	766	0	0	0	0	Net debt/EBITDA	NM	6.6	NM	NM	NM
Payables	28	30	21	25	115	Sales/Assets (x)	1.2	0.2	0.4	0.5	1.2
Other short term liabilities	13	4	10	13	62	Assets/Equity (x)	NM	NM	1.1	1.1	1.1
Current liabilities	807	34	30	39	178	Interest cover (x)	NM	NM	114.5	328.5	5,647.5
Long-term debt	-	-	-	-	-	Operating leverage	(87.0%)	(209.0%)	(55.0%)	418.0%	390.2%
Other long term liabilities	1	4	4	4	4	Tax rate	(1.0%)	(0.0%)	0.0%	0.0%	0.0%
Total liabilities	808	38	34	43	182	Revenue y/y Growth	67.7%	(34.5%)	242.1%	45.5%	437.2%
Shareholders' equity	(664)	452	494	579	1,729	EBITDA y/y Growth	(65.5%)	90.0%	(138.3%)	186.8%	1619.2%
Minority interests	-	-	-	-	-	EPS y/y Growth	-	-	(196.5%)	100.7%	1255.1%
Total liabilities & equity	144	490	529	622	1,911	Valuation	FY24A	FY25A	FY26E	FY27E	FY28E
BVPS	-	-	-	-	-	P/E (x)	-	NM	81.7	40.7	3.0
y/y Growth	-	-	-	-	-	P/BV (x)	-	-	-	-	-
Net debt/(cash)	640	(393)	(386)	(437)	(1,275)	EV/EBITDA (x)	NM	NM	129.6	44.4	1.8
						Dividend Yield	-	-	-	-	-

Source: Company reports and J.P. Morgan estimates.

Note: \$ in millions (except per-share data). Fiscal year ends Dec. o/w - out of which

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XtalPi - H (2228.HK, 2228 HK) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.

Insilico - H (3696.HK, 3696 HK) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.

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